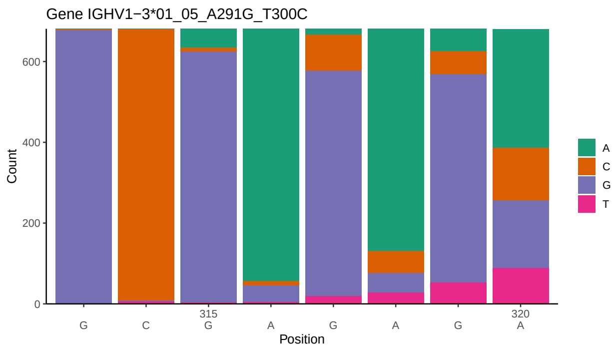
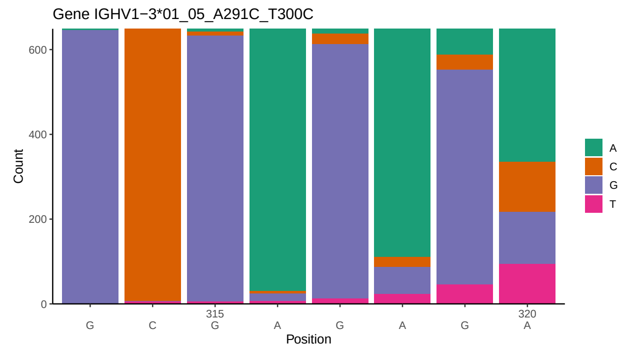
OGRDBstats Report

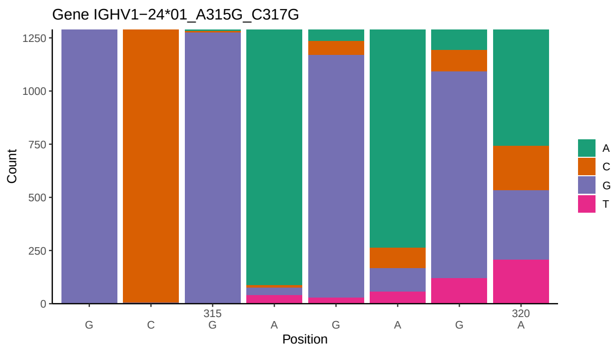
Contents

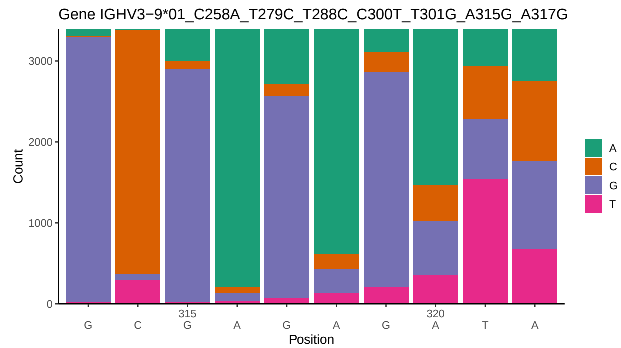
1 Novel sequence analysis

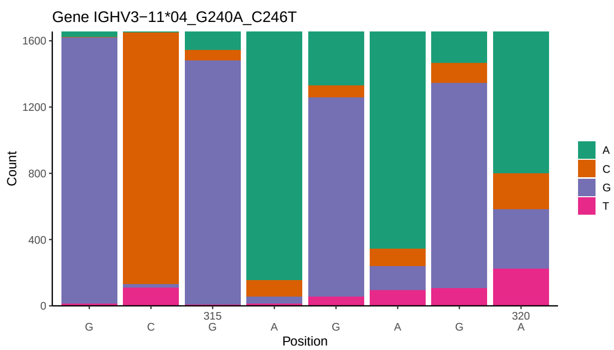
1.1 End-nucleotide composition

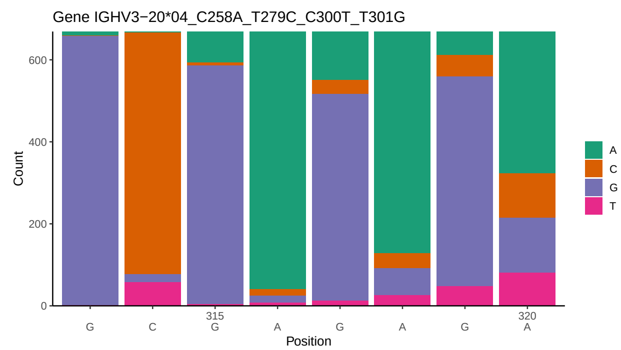


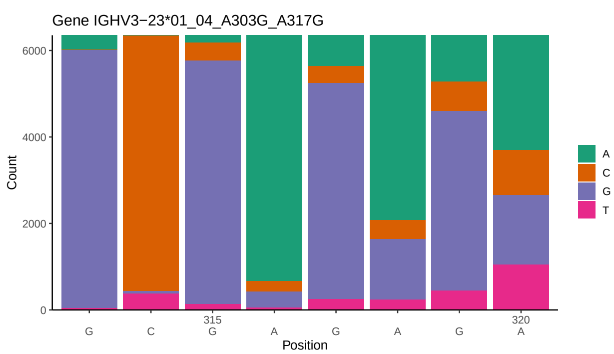


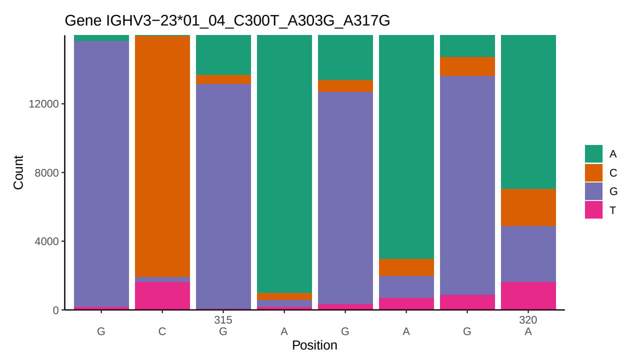


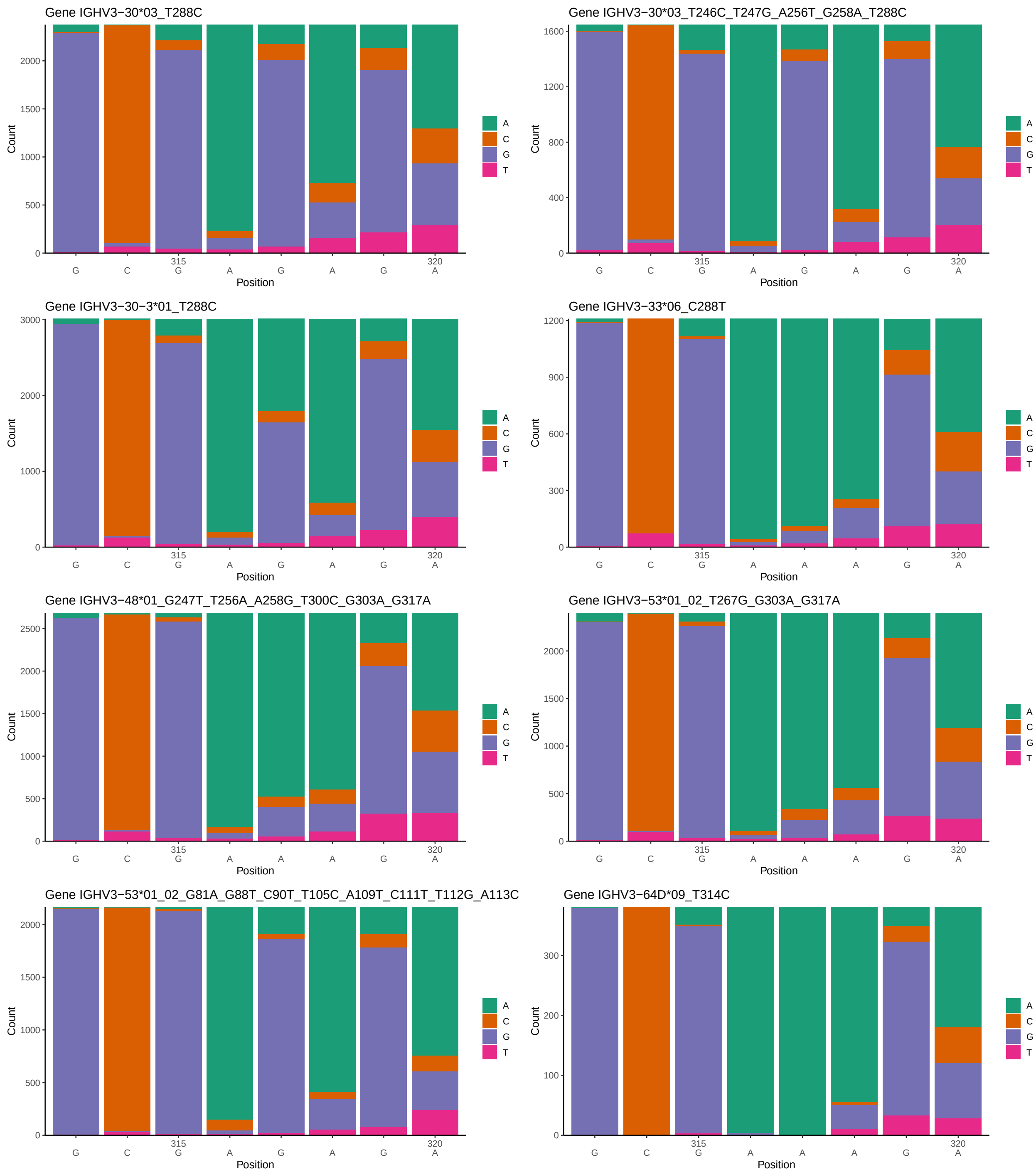








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Gene IGHV3−33*06_C288T
Gene IGHV3−48*01_G247T_T256A_A258G_T300C_G303A_G317A
Gene IGHV3−53*01_02_T267G_G303A_G317A
Gene IGHV3−53*01_02_G81A_G88T_C90T_T105C_A109T_C111T_T112G_A113C
Gene IGHV3−64D*09_T314C
Count
Position
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320
A
C
G
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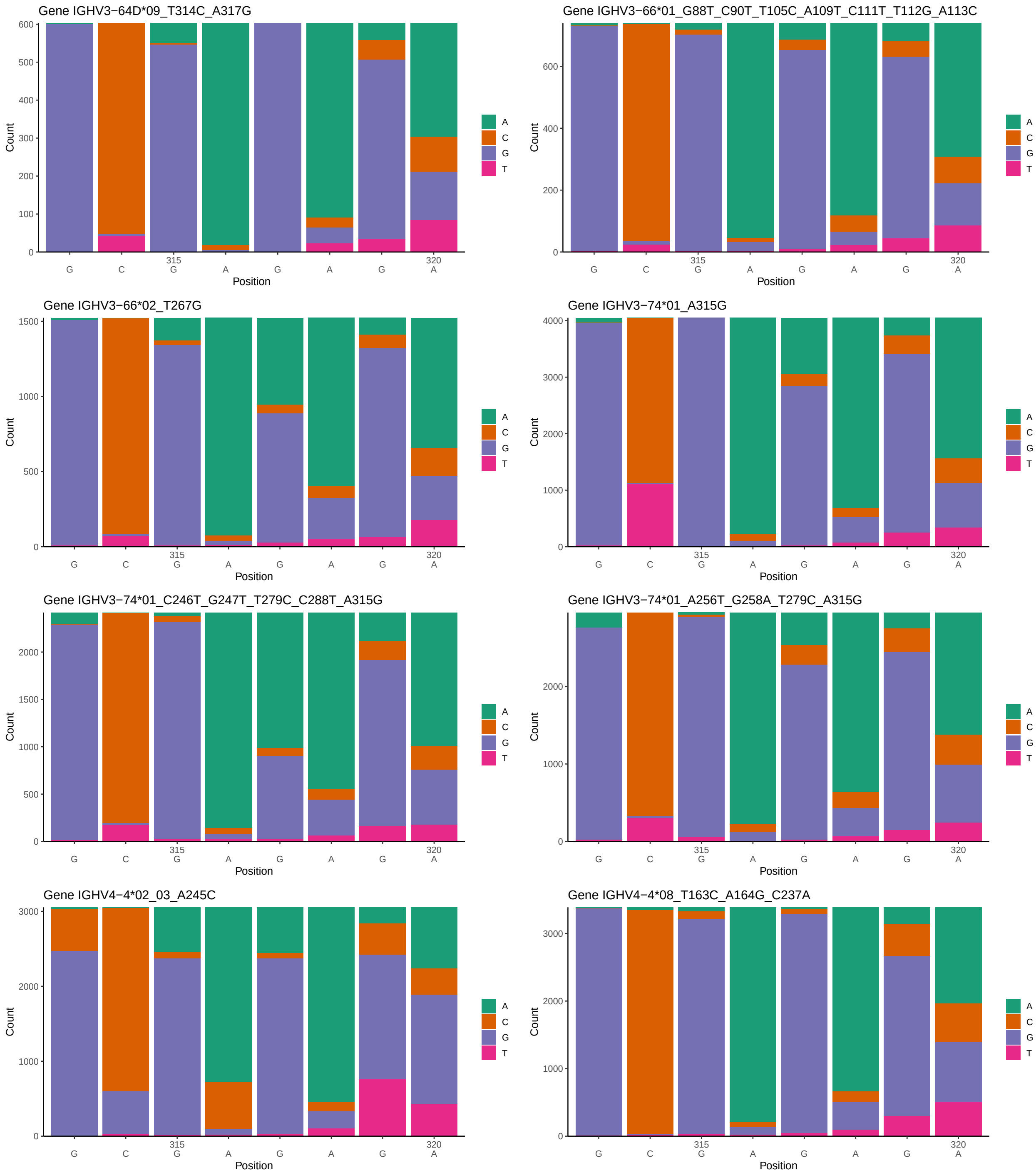

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Gene IGHV3−74*01_A315G
Gene IGHV3−74*01_C246T_G247T_T279C_C288T_A315G
Gene IGHV3−74*01_A256T_G258A_T279C_A315G
Gene IGHV4−4*02_03_A245C
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C
G
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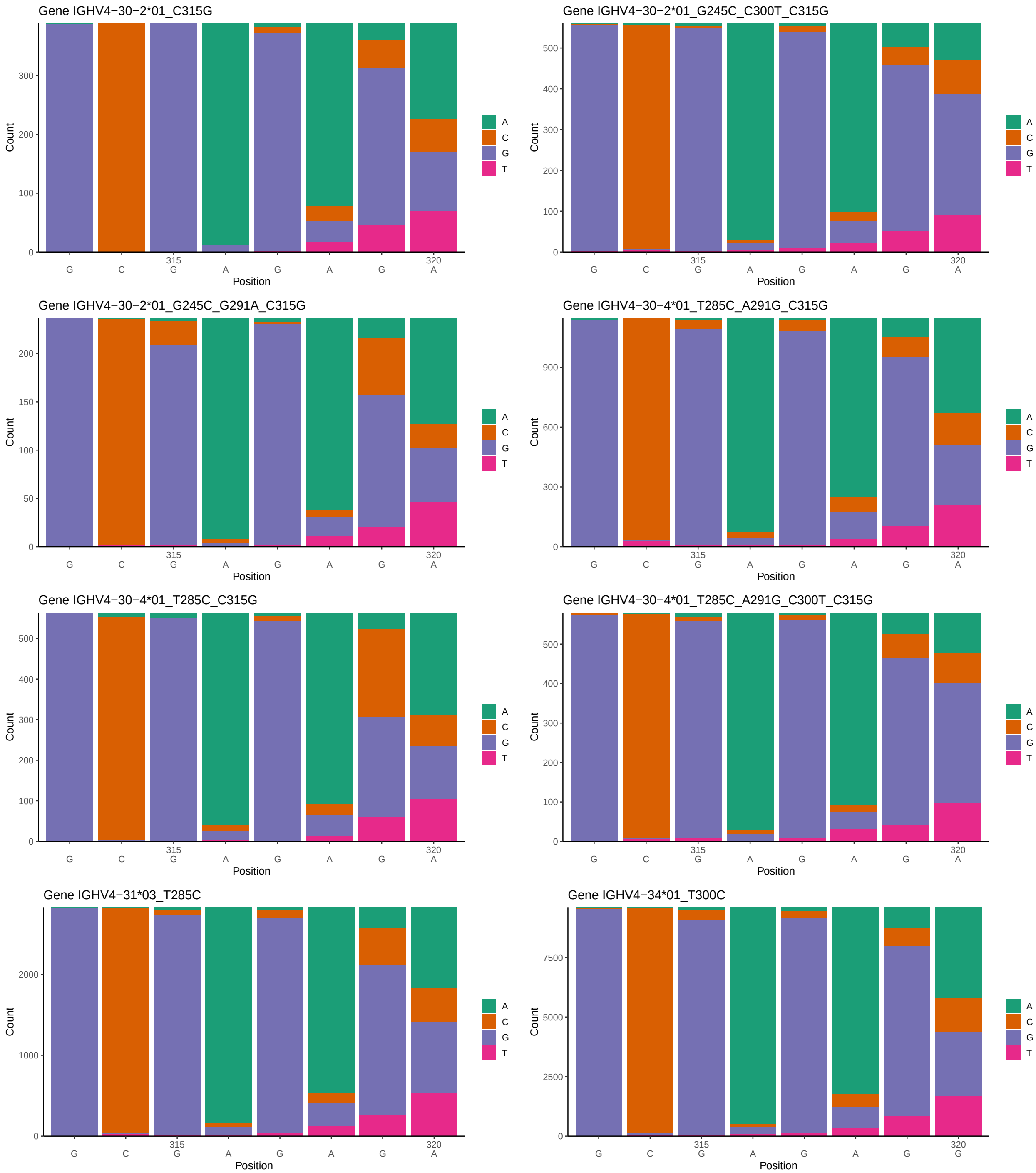
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Gene IGHV4−30−4*01_T285C_A291G_C315G
Gene IGHV4−30−4*01_T285C_C315G
Gene IGHV4−30−4*01_T285C_A291G_C300T_C315G
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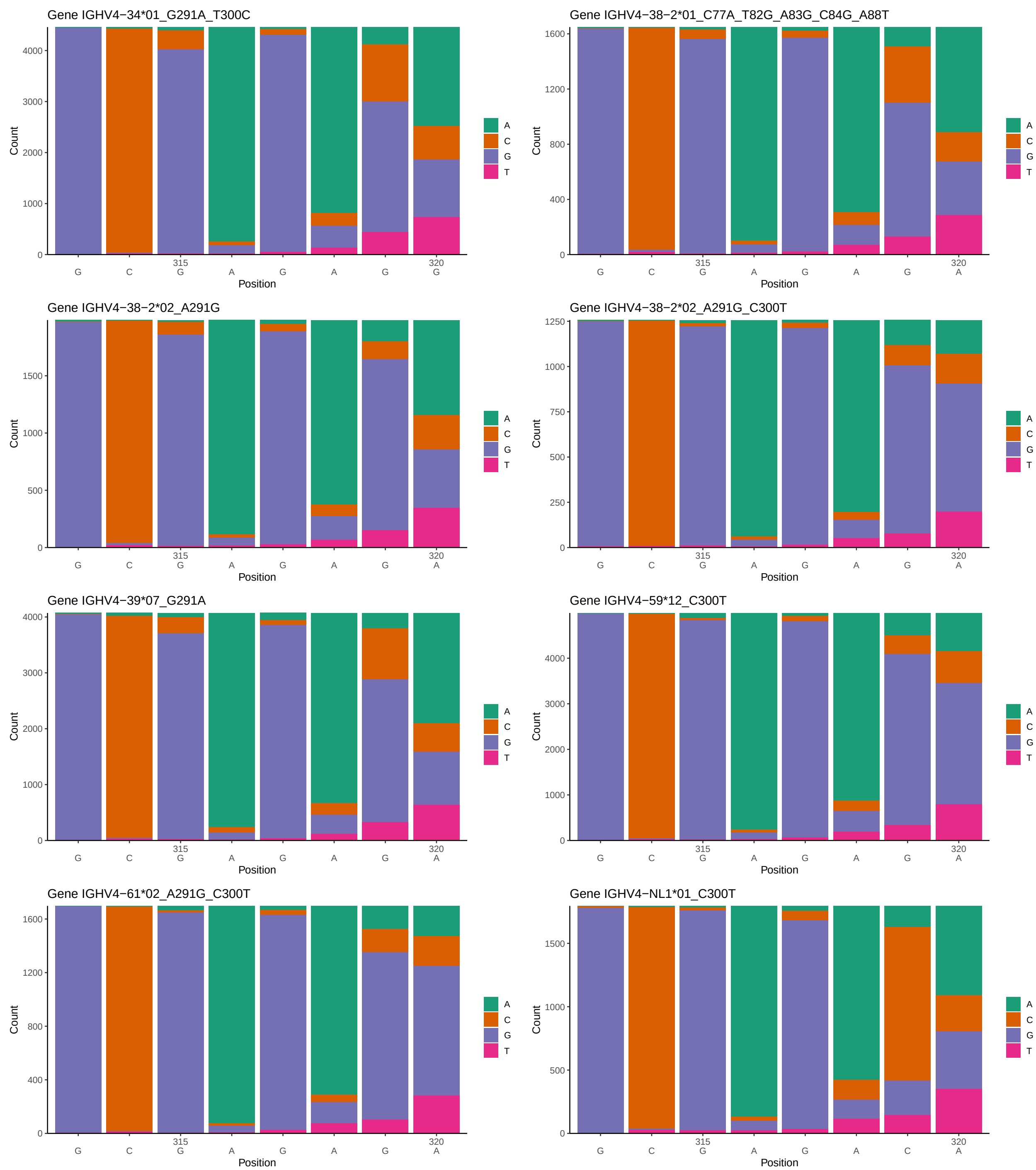
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A C G T
Gene IGHV4−38−2*02_A291G
Count
Position
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A C G T
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Count
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A C G T
Gene IGHV4−39*07_G291A
Count
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G C 315 G A G A G 320 A
A C G T
Gene IGHV4−59*12_C300T
Count
Position
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A C G T
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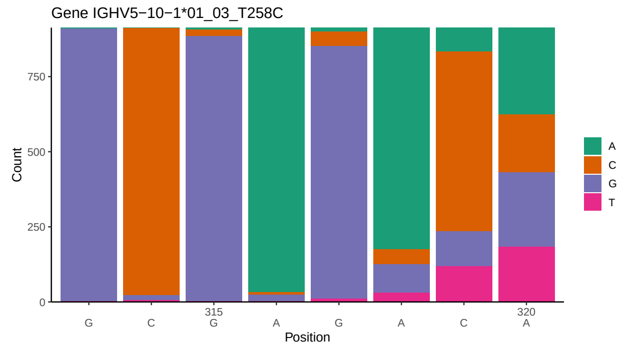

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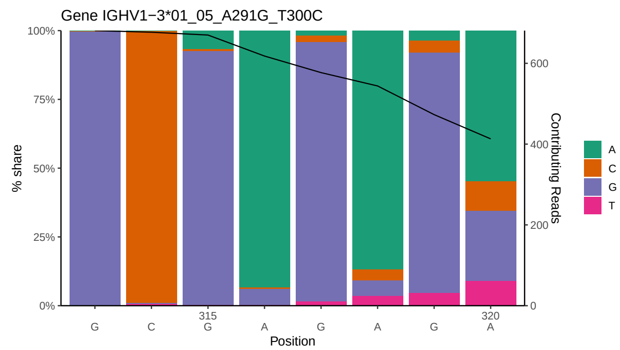

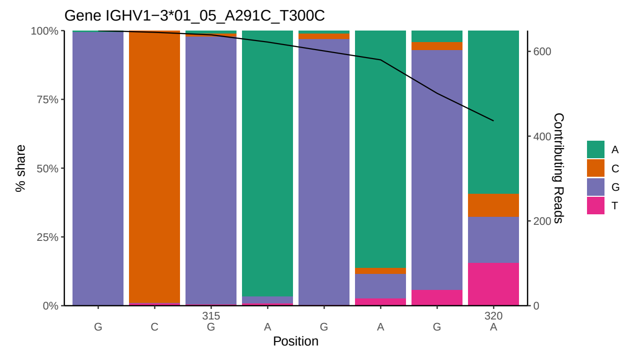

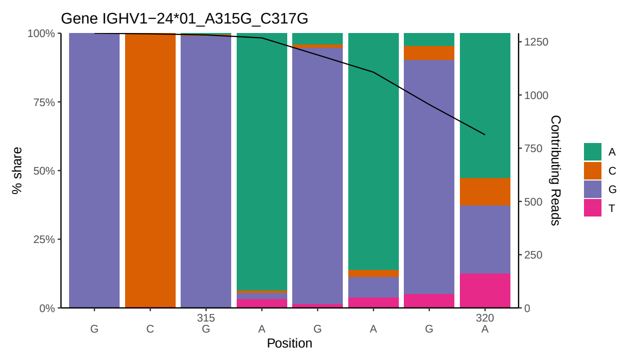

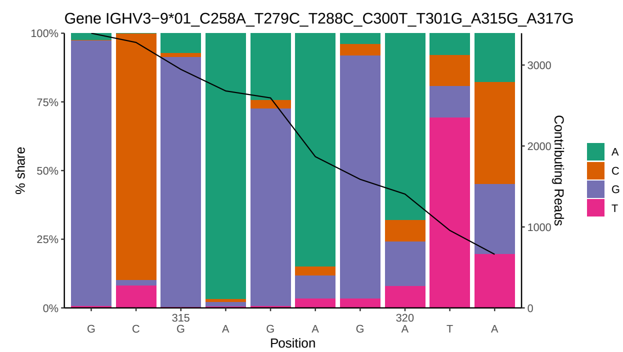

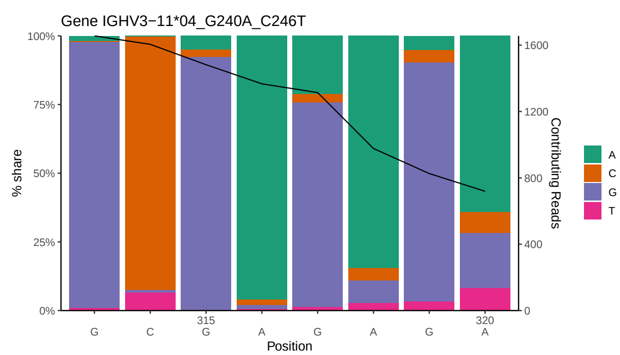

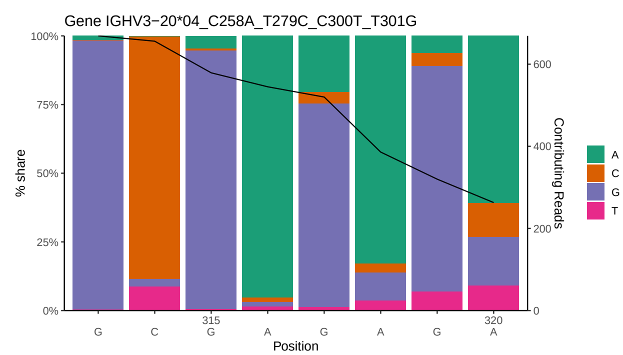

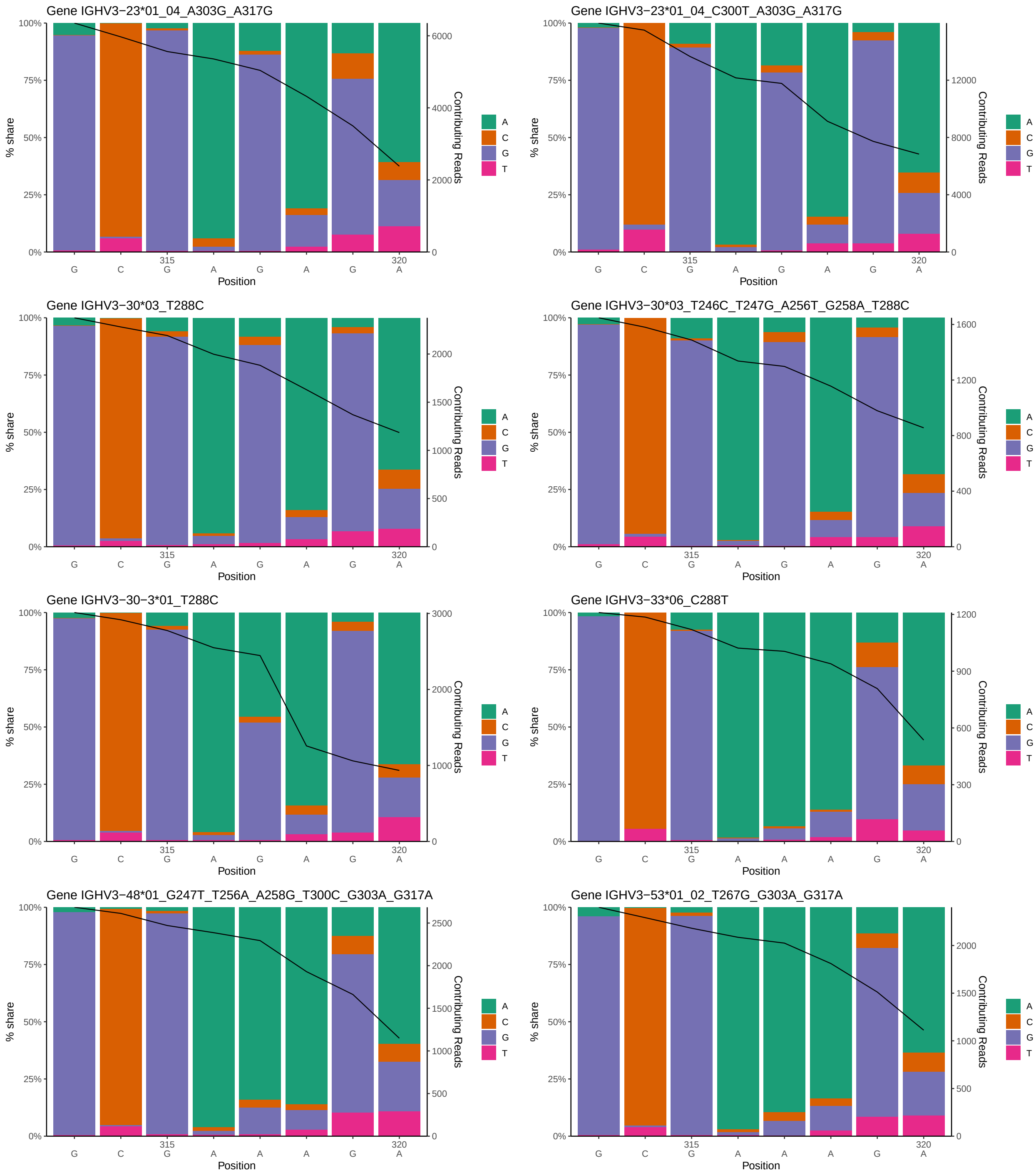
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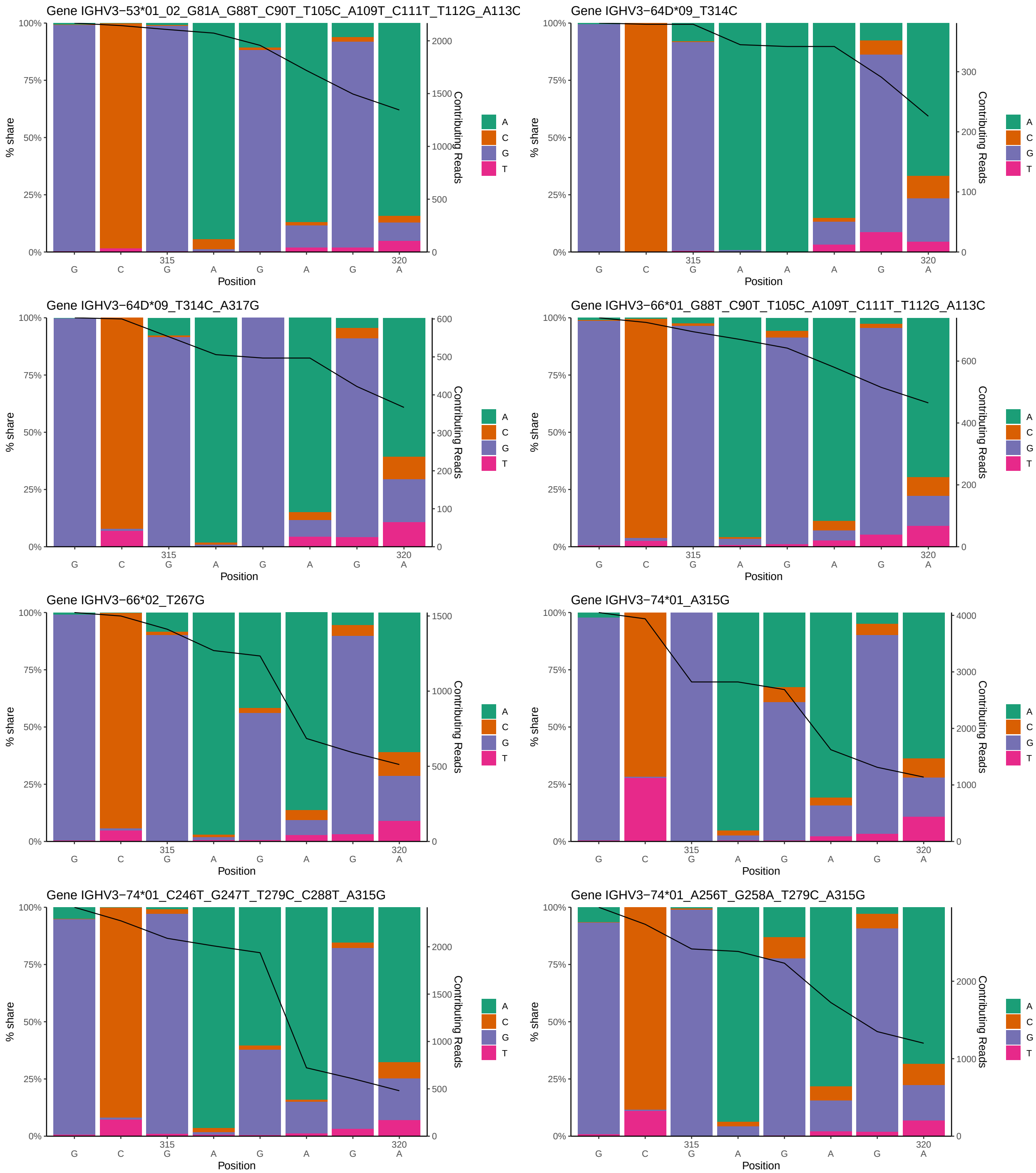
Gene IGHV3−30*03_T288C
Gene IGHV3−30*03_T246C_T247G_A256T_G258A_T288C
Gene IGHV3−30−3*01_T288C
Gene IGHV3−33*06_C288T
Gene IGHV3−48*01_G247T_T256A_A258G_T300C_G303A_G317A
Gene IGHV3−53*01_02_T267G_G303A_G317A
% share
Contributing Reads
Position
G C G A G A G A
315
320
A
C
G
T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Gene IGHV3−64D*09_T314C
Gene IGHV3−64D*09_T314C_A317G
Gene IGHV3−66*01_G88T_C90T_T105C_A109T_C111T_T112G_A113C
Gene IGHV3−66*02_T267G
Gene IGHV3−74*01_A315G
Gene IGHV3−74*01_C246T_G247T_T279C_C288T_A315G
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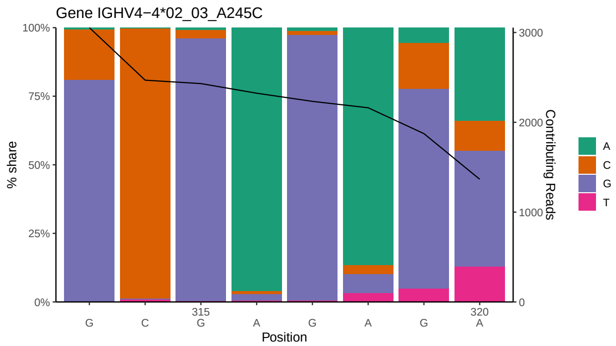
% share
Contributing Reads
Position
A
C
G
T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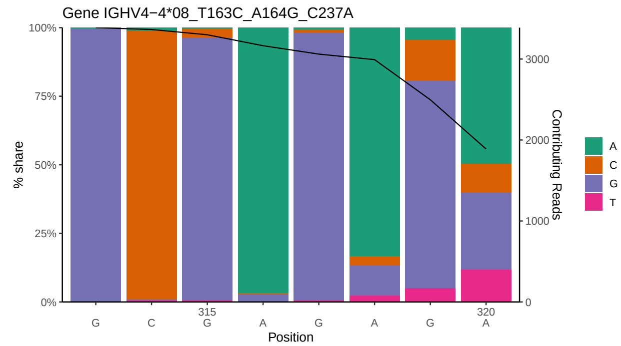
A
C
G
T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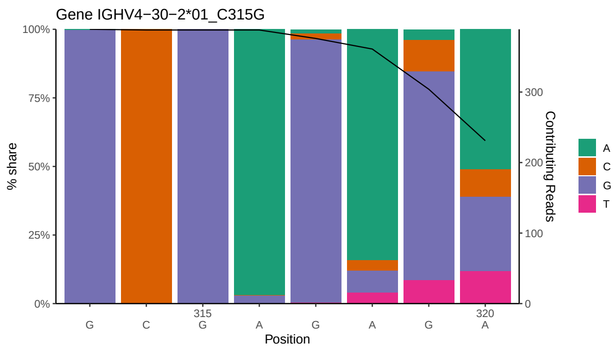
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C
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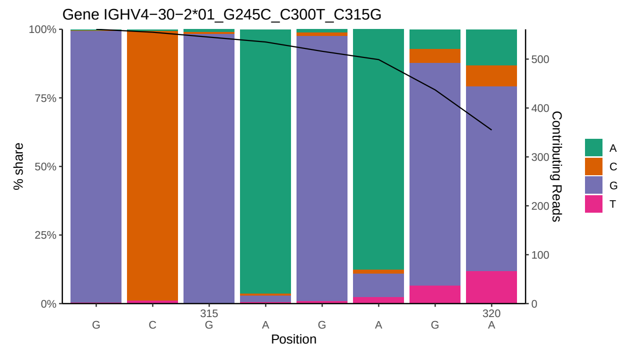
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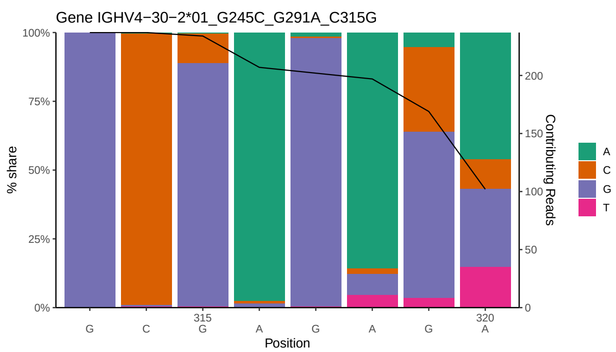
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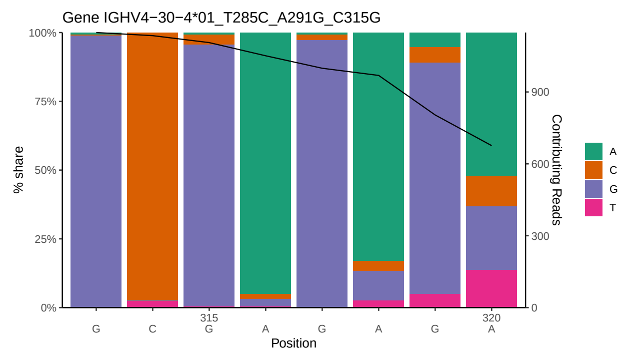
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C
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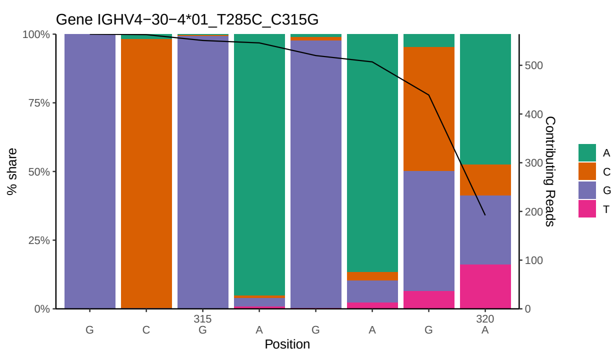
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A
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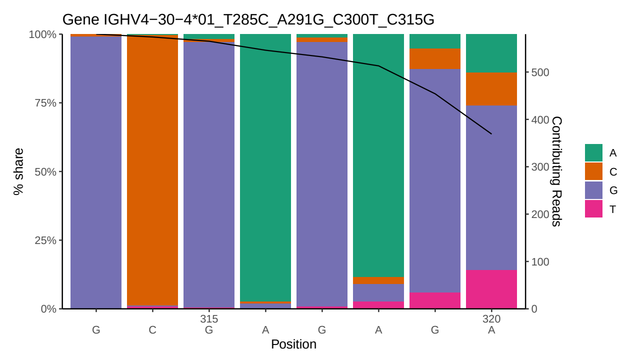
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A
C
G
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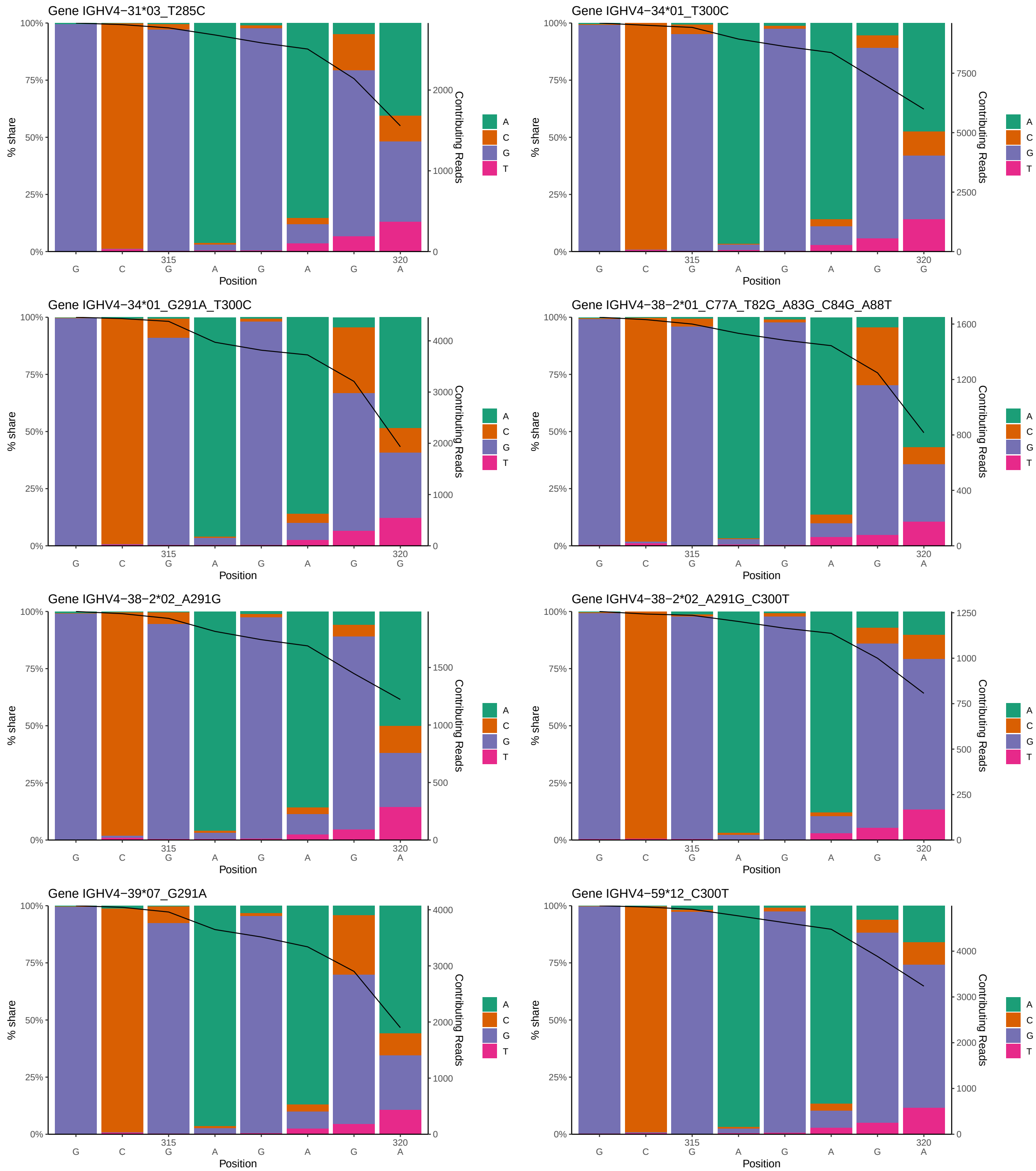
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Gene IGHV4−34*01_G291A_T300C
Gene IGHV4−38−2*01_C77A_T82G_A83G_C84G_A88T
Gene IGHV4−38−2*02_A291G
Gene IGHV4−38−2*02_A291G_C300T
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Position
A
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G
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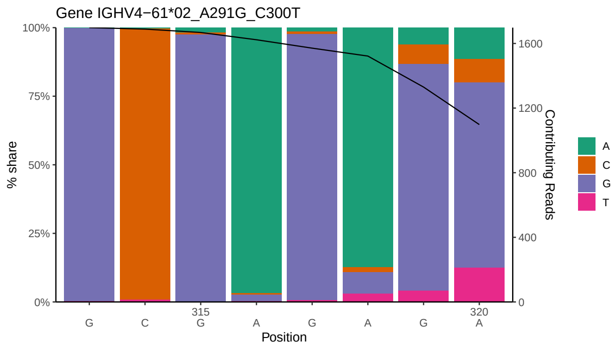
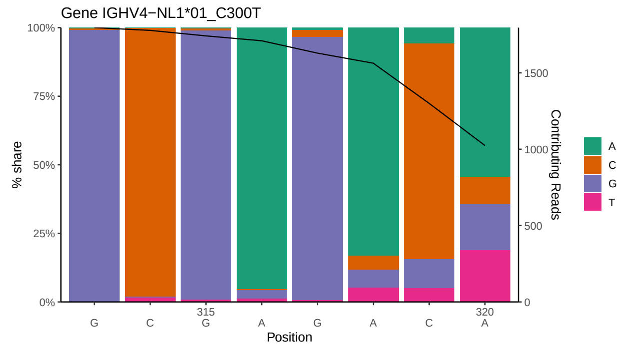
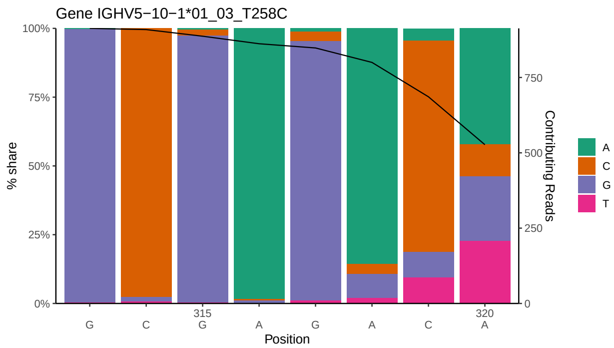
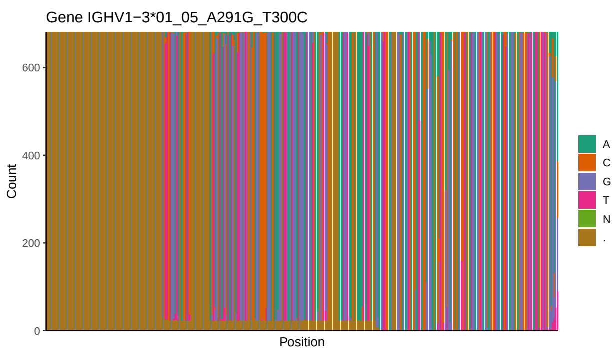
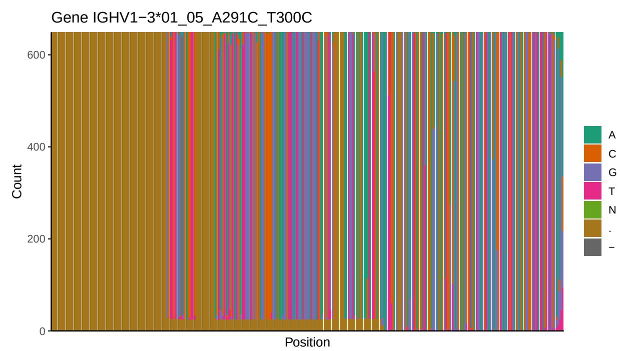


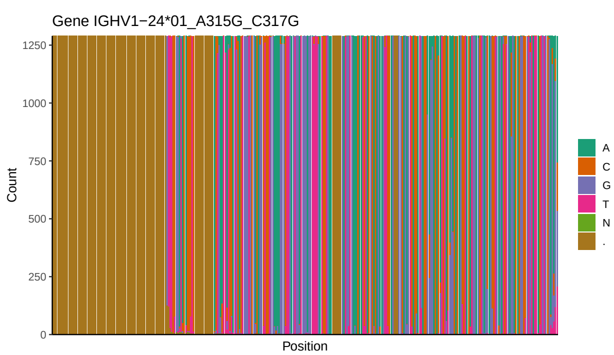


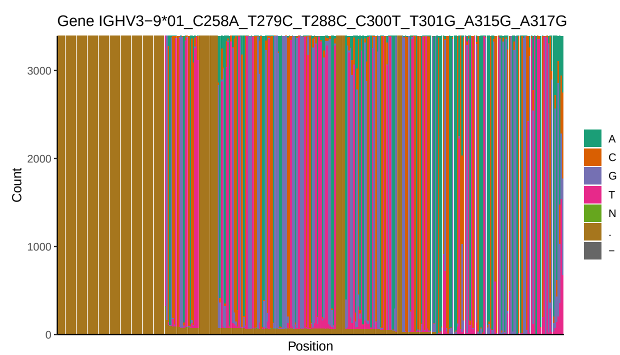


1.3 Whole-sequence composition of each assigned read









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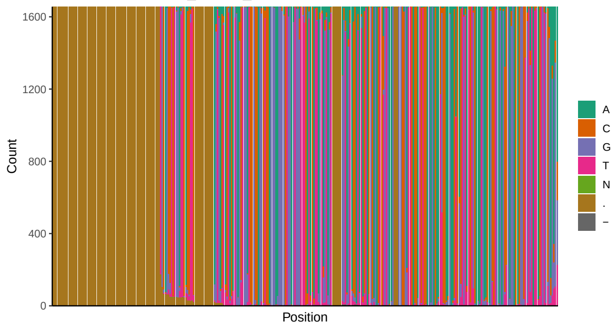

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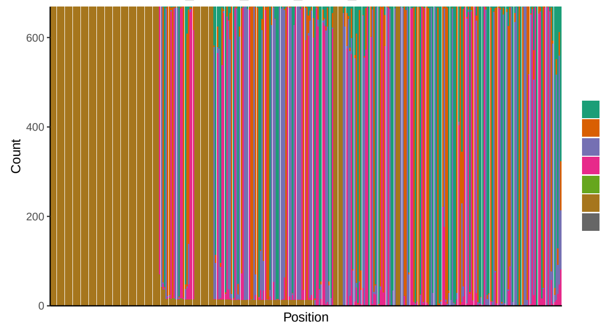

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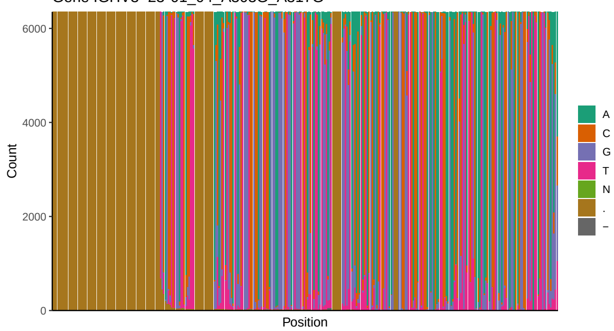

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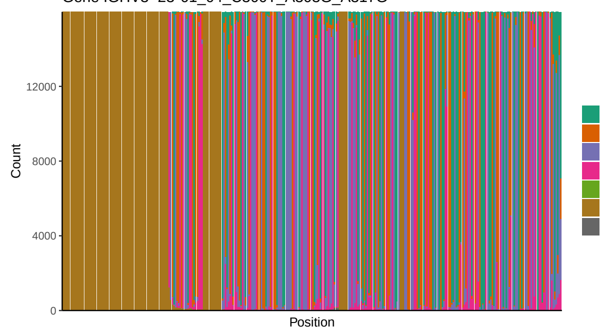

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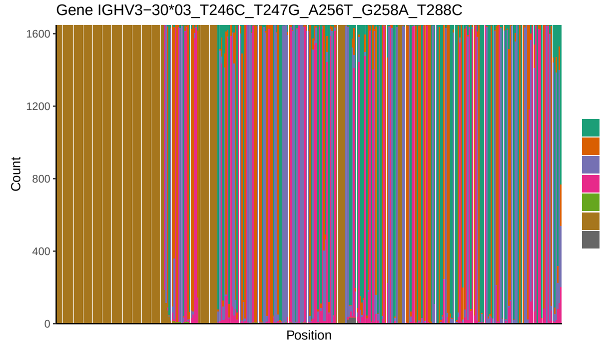
Gene IGHV3−30*03_T246C_T247G_A256T_G258A_T288C



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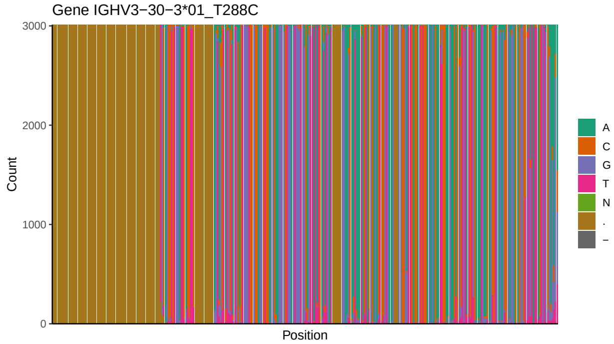

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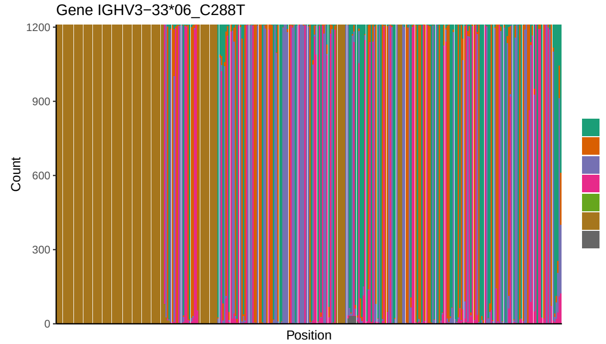

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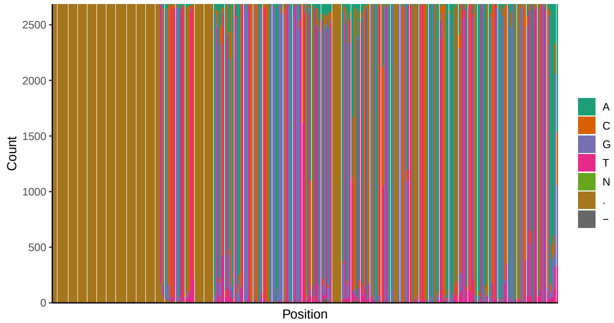

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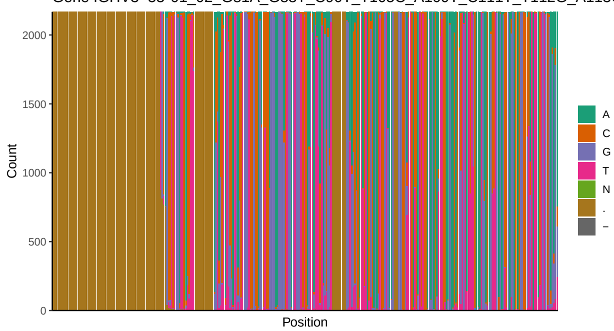

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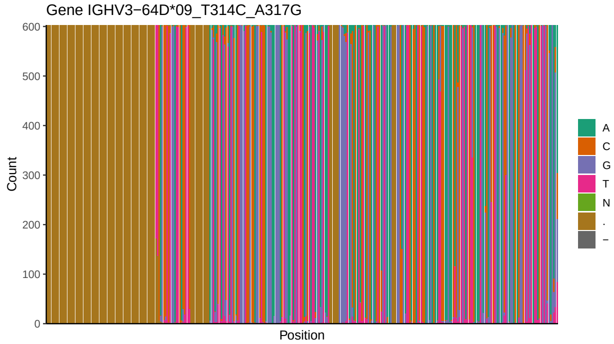

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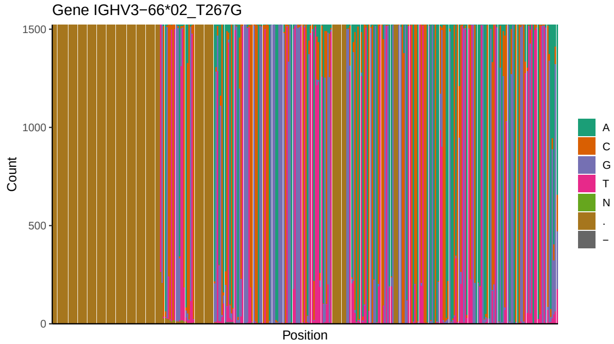

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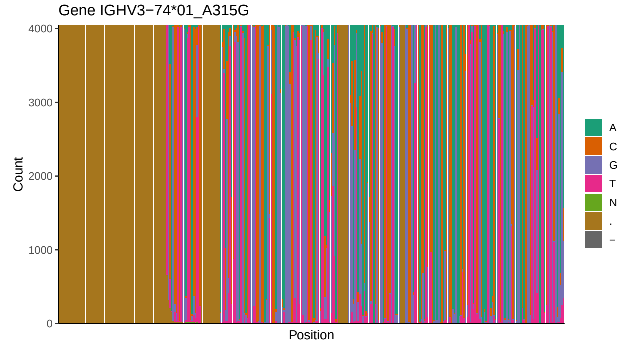

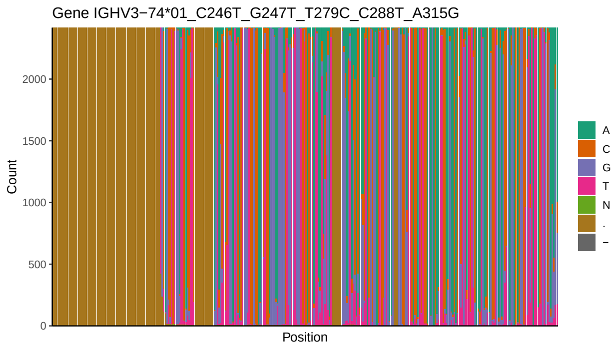
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.
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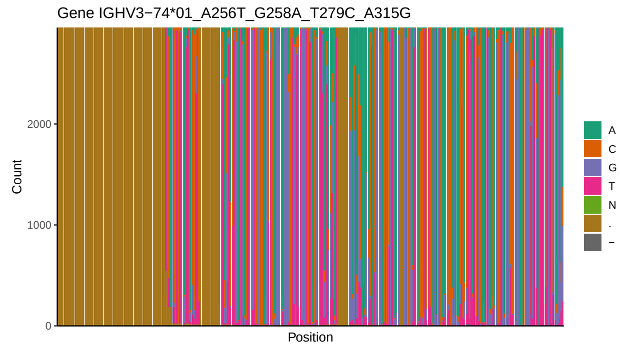
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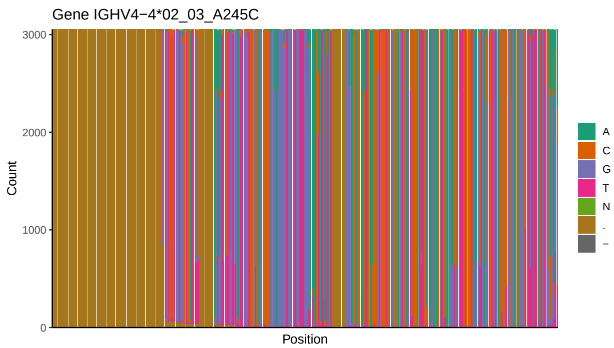
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Count
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1000
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3000
Position
A
C
G
T
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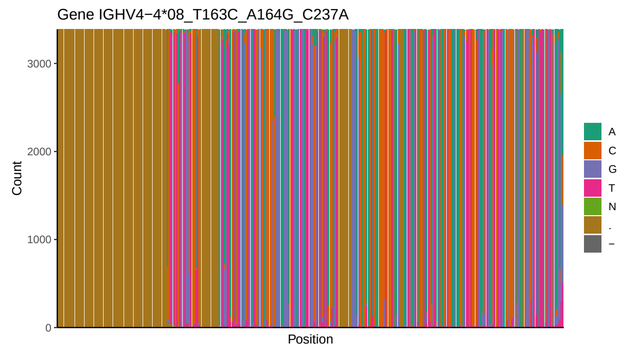
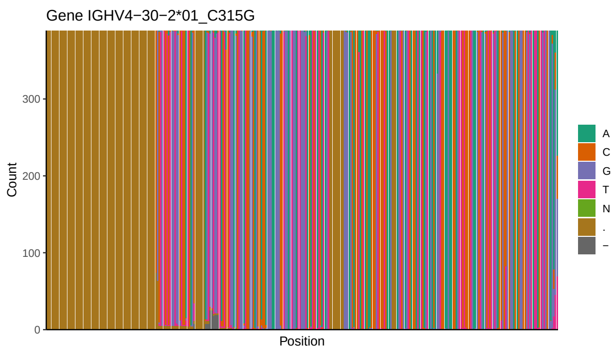
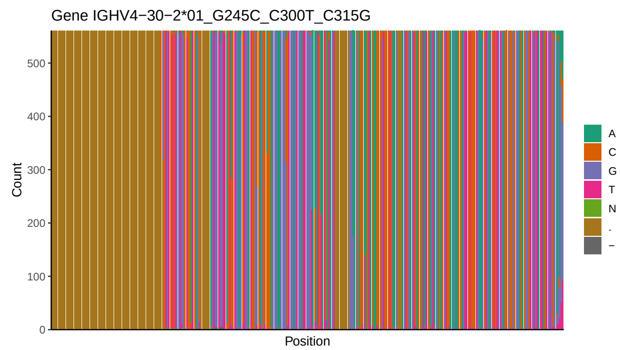
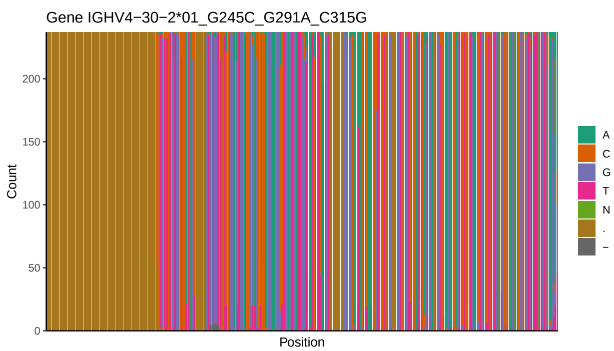
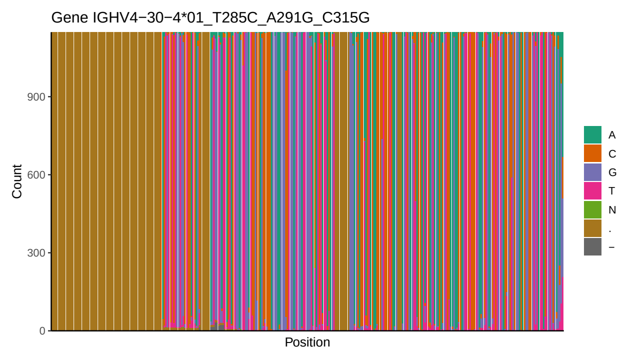
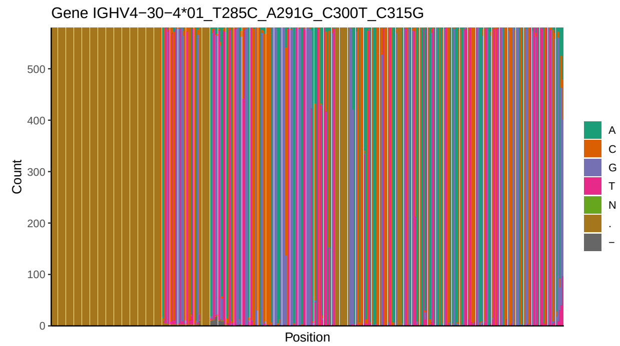
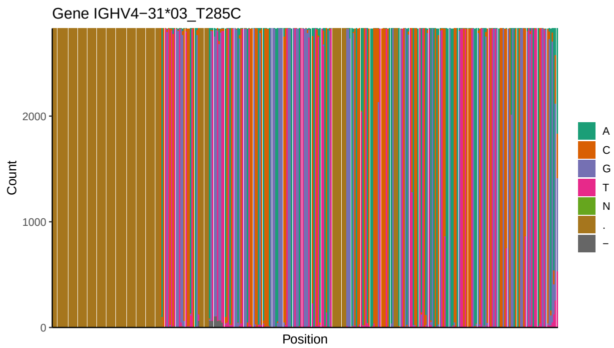
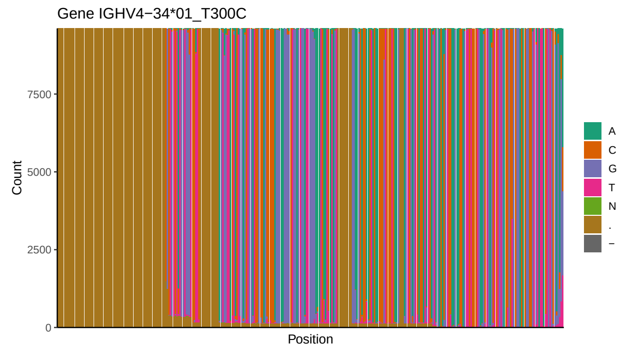
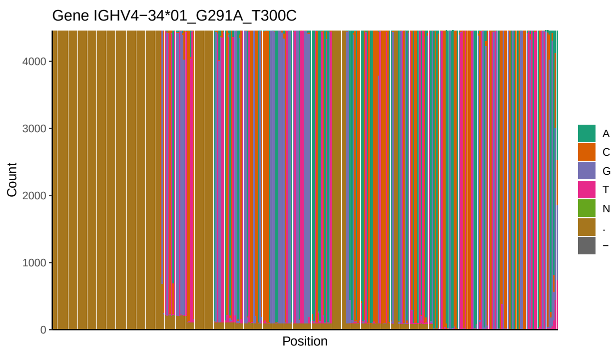
.
−

Gene IGHV4−4*08_T163C_A164G_C237A
Count
0
1000
2000
3000
Position
A
C
G
T
N
.
−

Gene IGHV4−30−2*01_C315G
Count
0
100
200
300
Position
A
C
G
T
N
.
−

Gene IGHV4−30−2*01_G245C_C300T_C315G
Count
0
100
200
300
400
500
Position
A
C
G
T
N
.
−

Gene IGHV4−30−2*01_G245C_G291A_C315G
Count
0
50
100
150
200
Position
A
C
G
T
N
.
−

Gene IGHV4−30−4*01_T285C_A291G_C315G
Count
0
300
600
900
Position
A
C
G
T
N
.
−

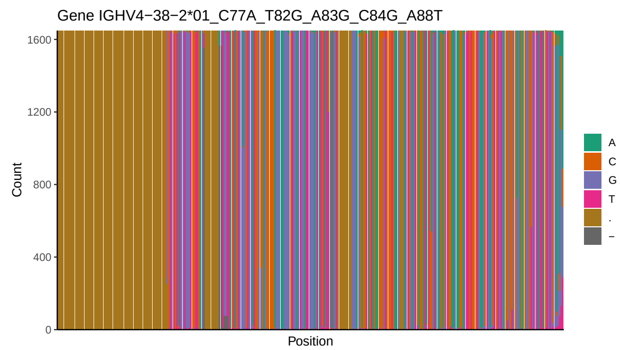

Gene IGHV4−30−4*01_T285C_C315G
Count
0
100
200
300
400
500
Position
A
C
G
T
N
.
−

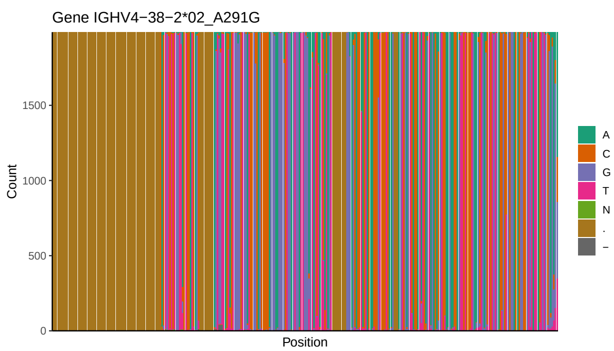

Gene IGHV4−30−4*01_T285C_A291G_C300T_C315G
Count
0
100
200
300
400
500
Position
A
C
G
T
N
.
−

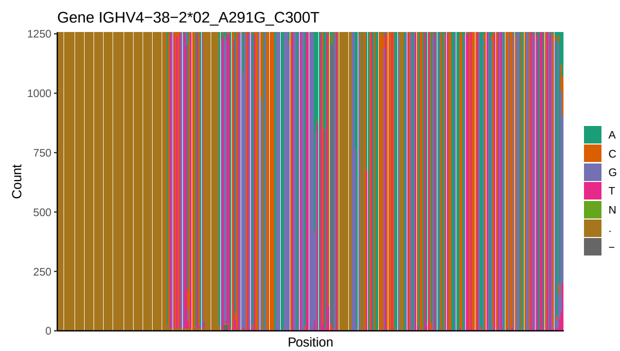

Gene IGHV4−31*03_T285C
Count
0
1000
2000
Position
A
C
G
T
N
.
−

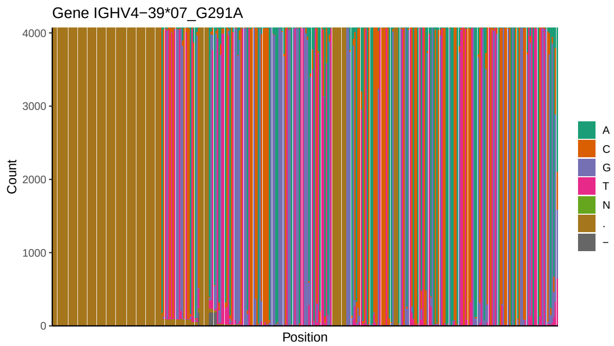

Gene IGHV4−34*01_T300C
Count
0
2500
5000
7500
Position
A
C
G
T
N
.
−

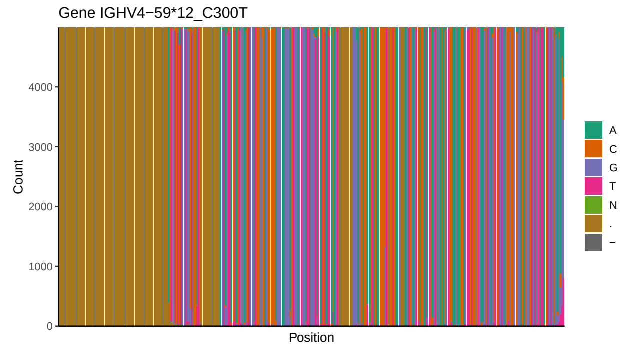

Gene IGHV4−34*01_G291A_T300C
Count
0
1000
2000
3000
4000
Position
A
C
G
T
N
.
−

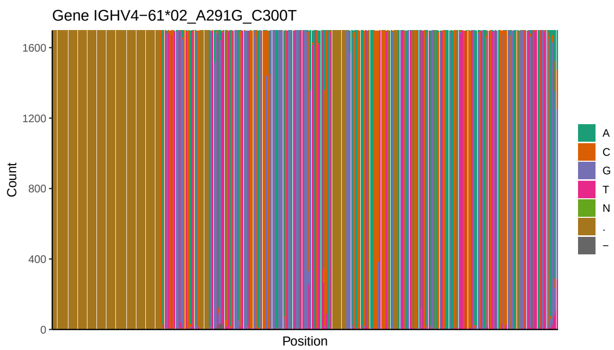

Gene IGHV4−38−2*01_C77A_T82G_A83G_C84G_A88T
Count
0
400
800
1200
1600
Position
A
C
G
T
.
−

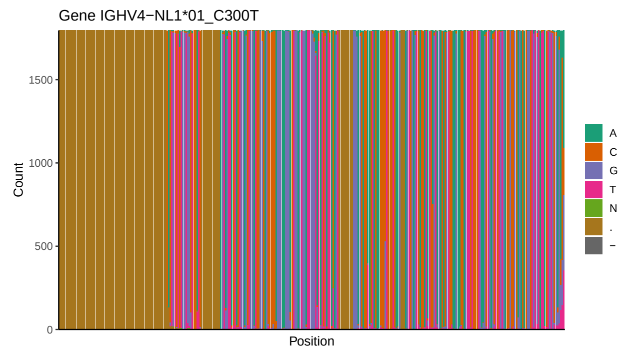

Gene IGHV4−38−2*02_A291G
Count
0
500
1000
1500
Position
A
C
G
T
N
.
−

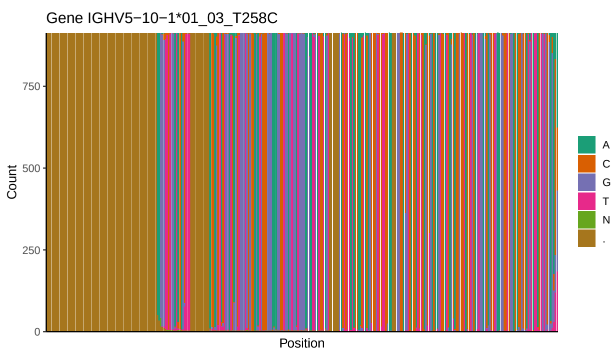

Gene IGHV4−38−2*02_A291G_C300T
Count
0
250
500
750
1000
1250
Position
A
C
G
T
N
.
−



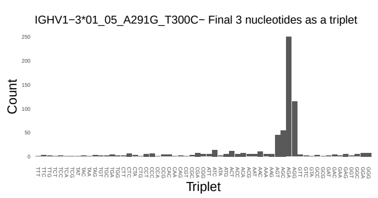


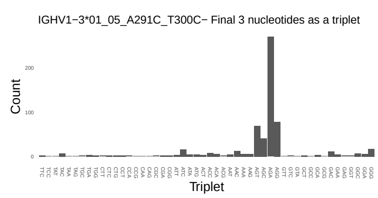


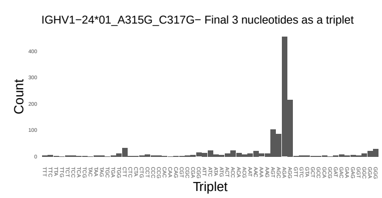


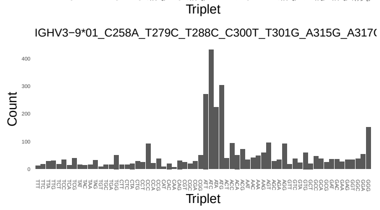


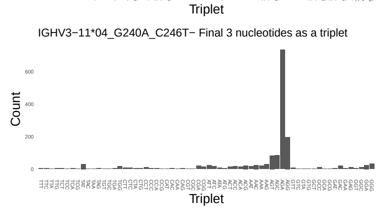
1.4 Final three nucleotides: frequency of each observed triplet

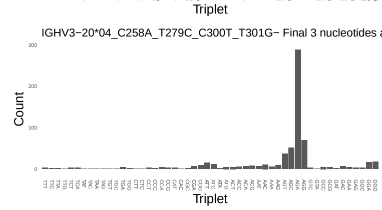


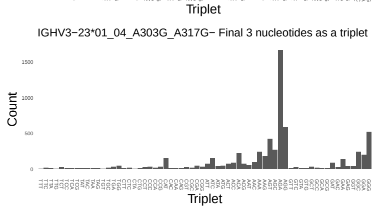


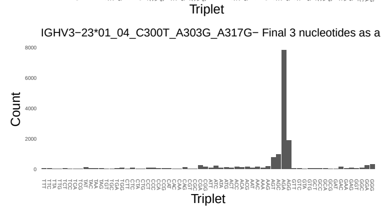


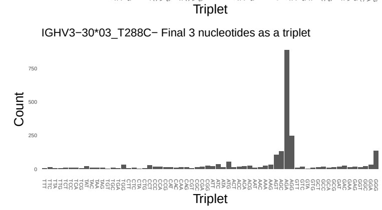


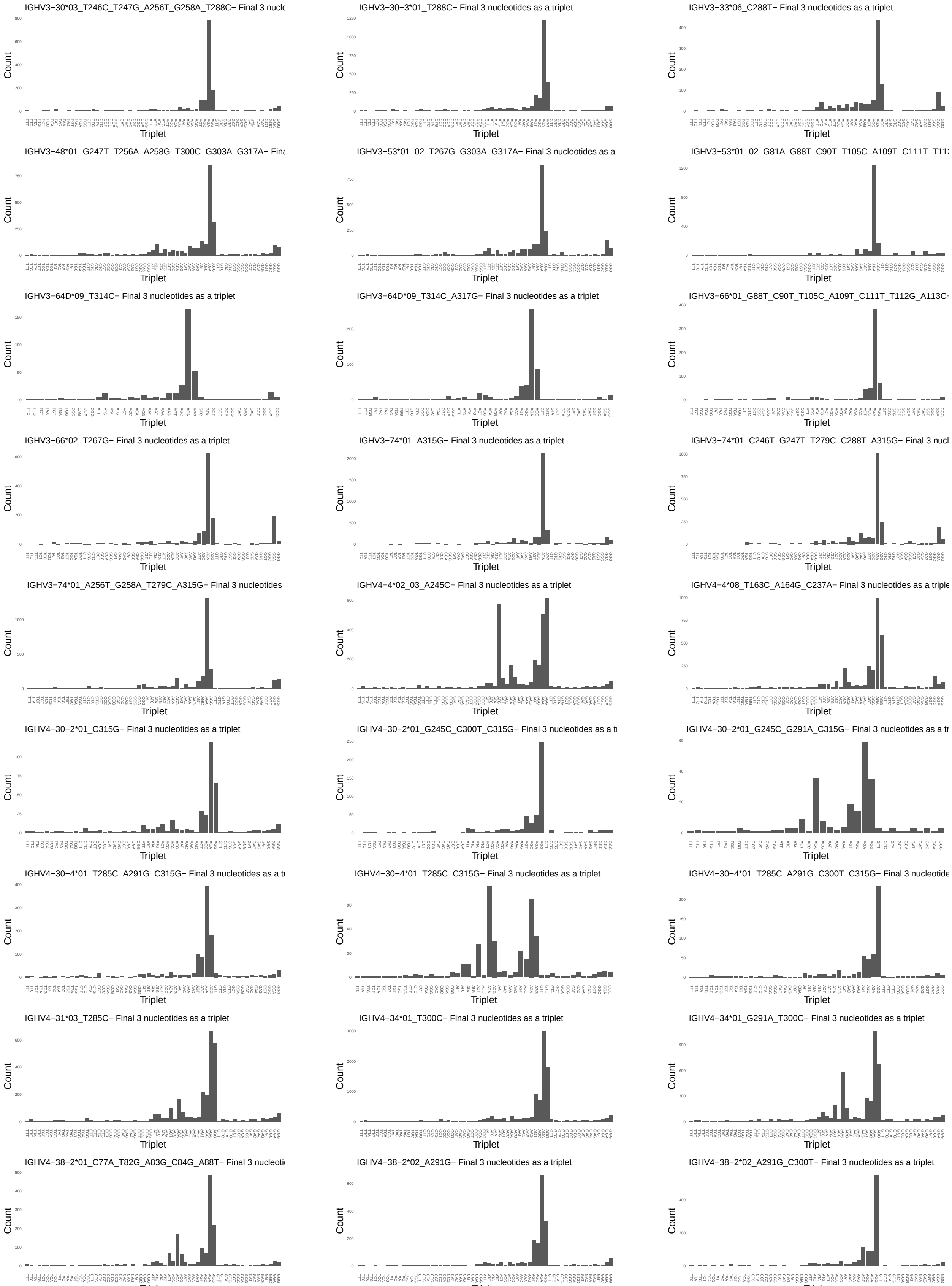


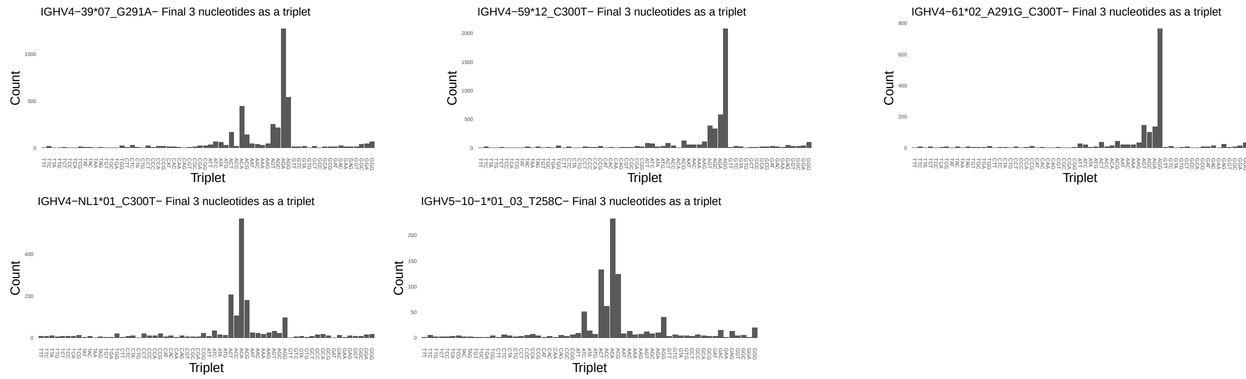




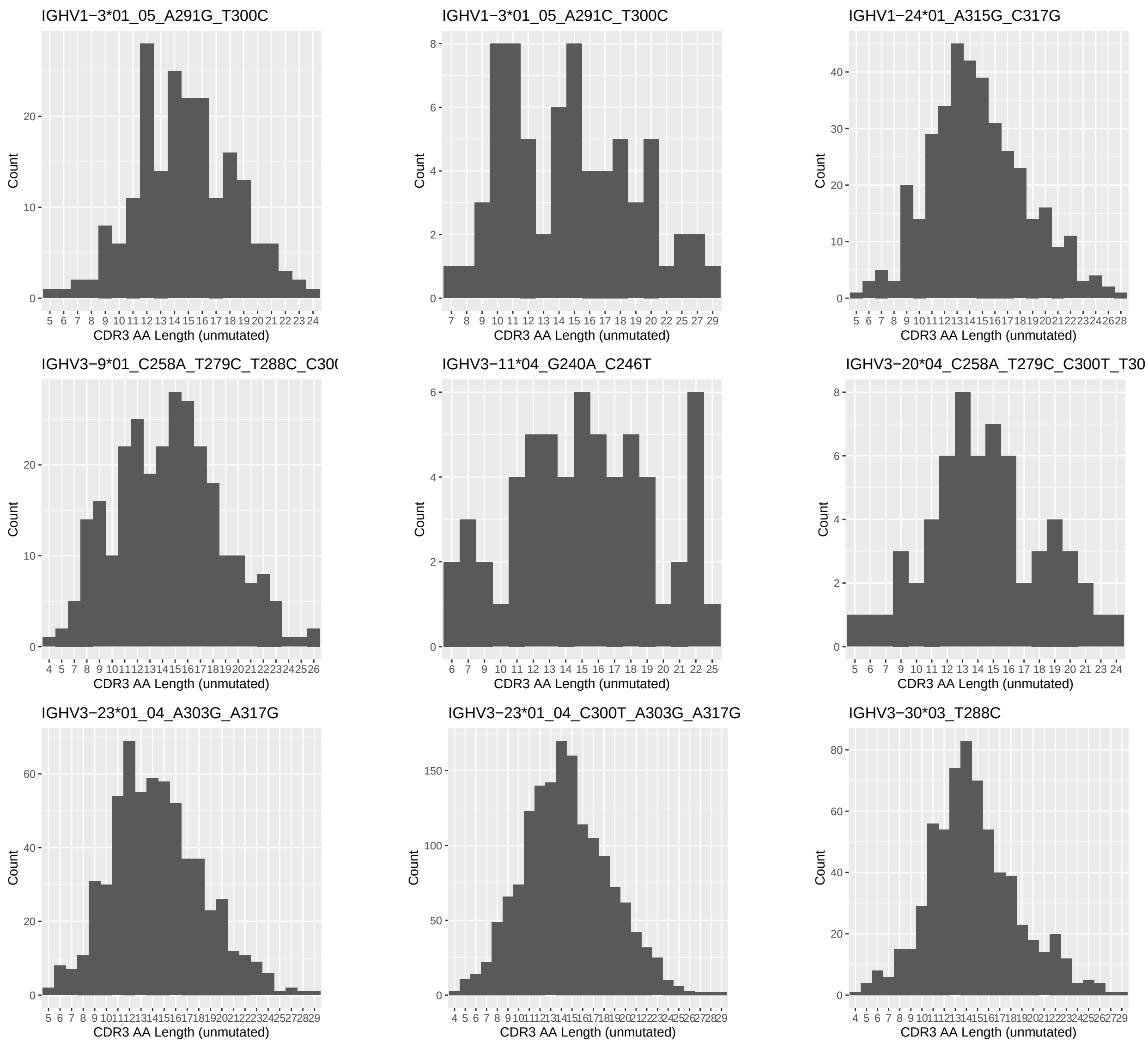


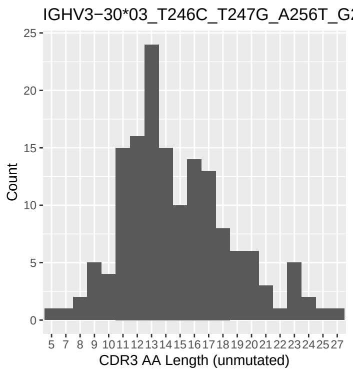
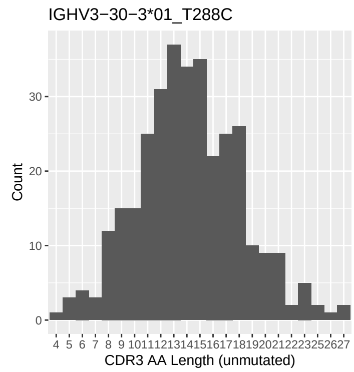
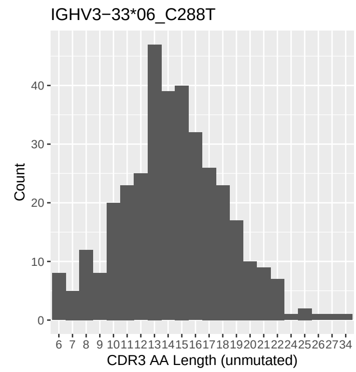
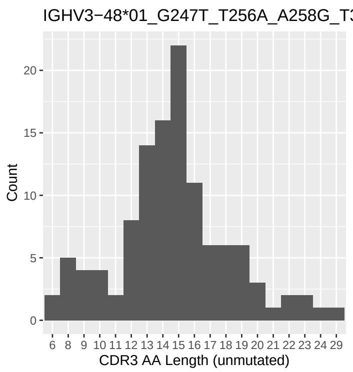
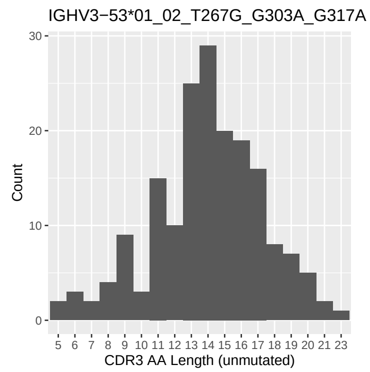
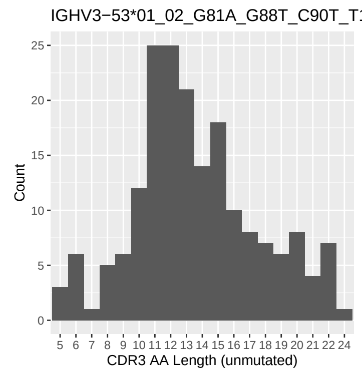
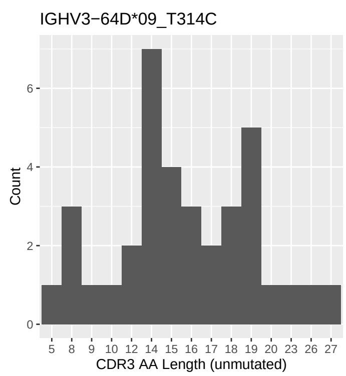
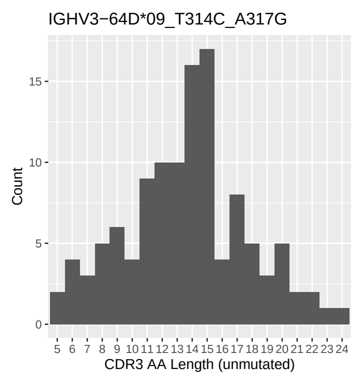
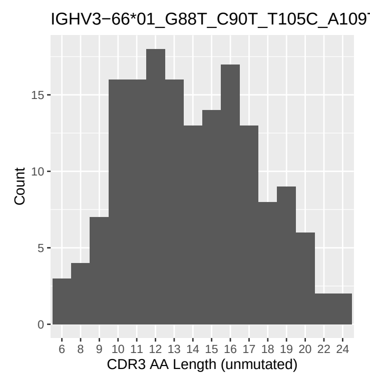
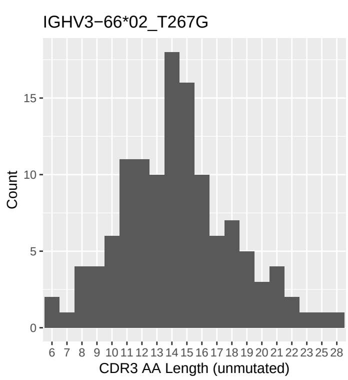
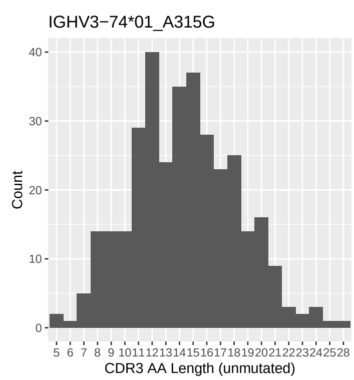
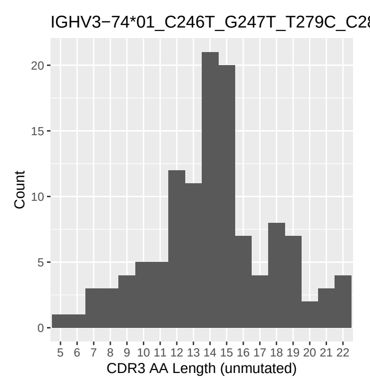


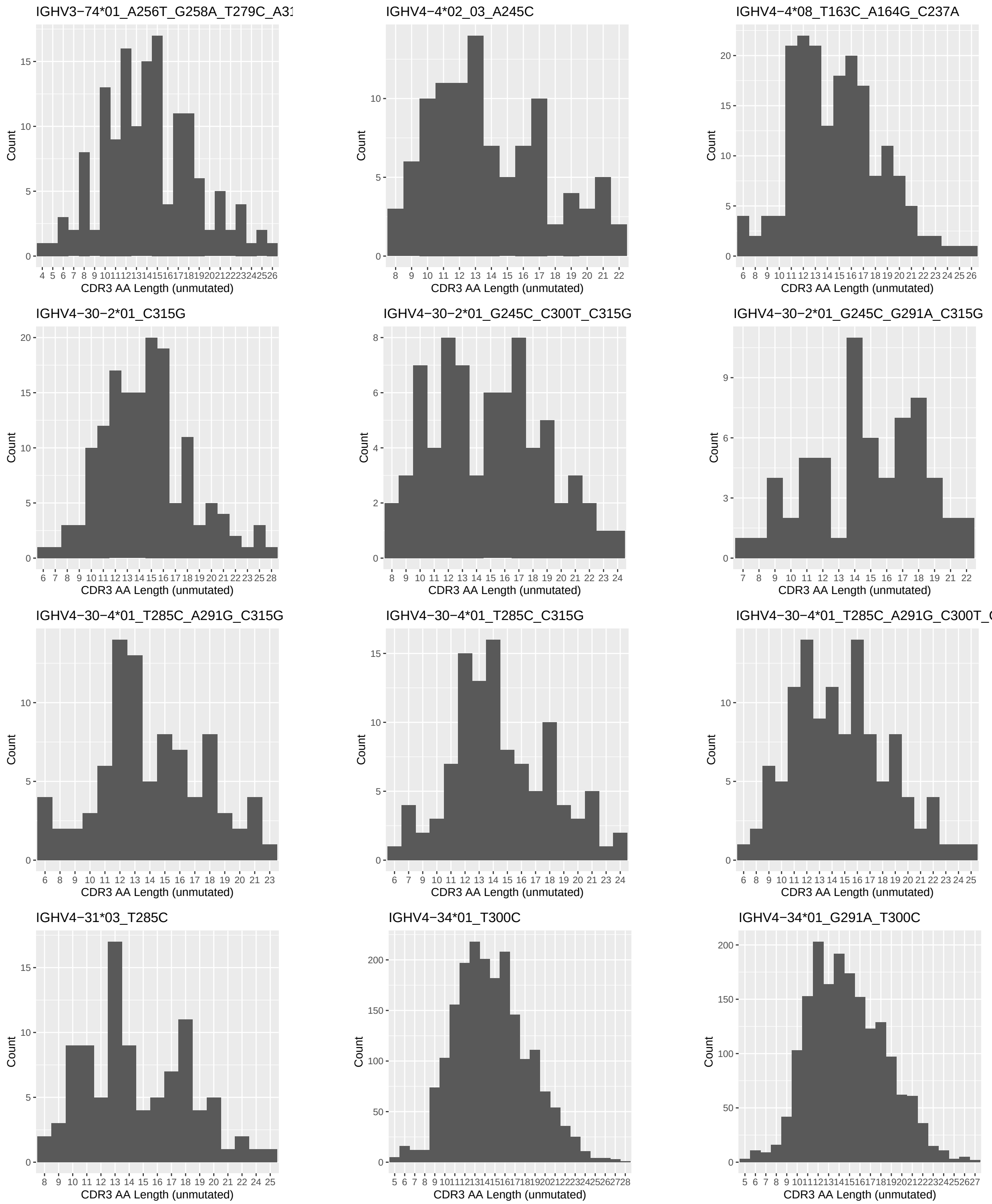

IGHV3−30*03_T246C_T247G_A256T_G258A_T288C− Final 3 nucle
IGHV3−30−3*01_T288C− Final 3 nucleotides as a triplet
IGHV3−33*06_C288T− Final 3 nucleotides as a triplet
IGHV3−48*01_G247T_T256A_A258G_T300C_G303A_G317A− Fina
IGHV3−53*01_02_T267G_G303A_G317A− Final 3 nucleotides as a
IGHV3−53*01_02_G81A_G88T_C90T_T105C_A109T_C111T_T11
IGHV3−64D*09_T314C− Final 3 nucleotides as a triplet
IGHV3−64D*09_T314C_A317G− Final 3 nucleotides as a triplet
IGHV3−66*01_G88T_C90T_T105C_A109T_C111T_T112G_A113C−
IGHV3−66*02_T267G− Final 3 nucleotides as a triplet
IGHV3−74*01_A315G− Final 3 nucleotides as a triplet
IGHV3−74*01_C246T_G247T_T279C_C288T_A315G− Final 3 nucl
IGHV3−74*01_A256T_G258A_T279C_A315G− Final 3 nucleotides
IGHV4−4*02_03_A245C− Final 3 nucleotides as a triplet
IGHV4−4*08_T163C_A164G_C237A− Final 3 nucleotides as a triple
IGHV4−30−2*01_C315G− Final 3 nucleotides as a triplet
IGHV4−30−2*01_G245C_C300T_C315G− Final 3 nucleotides as a t
IGHV4−30−2*01_G245C_G291A_C315G− Final 3 nucleotides as a tr
IGHV4−30−4*01_T285C_A291G_C315G− Final 3 nucleotides as a t
IGHV4−30−4*01_T285C_C315G− Final 3 nucleotides as a triplet
IGHV4−30−4*01_T285C_A291G_C300T_C315G− Final 3 nucleotide
IGHV4−31*03_T285C− Final 3 nucleotides as a triplet
IGHV4−34*01_T300C− Final 3 nucleotides as a triplet
IGHV4−34*01_G291A_T300C− Final 3 nucleotides as a triplet
IGHV4−38−2*01_C77A_T82G_A83G_C84G_A88T− Final 3 nucleoti
IGHV4−38−2*02_A291G− Final 3 nucleotides as a triplet
IGHV4−38−2*02_A291G_C300T− Final 3 nucleotides as a triplet
Count
Triplet

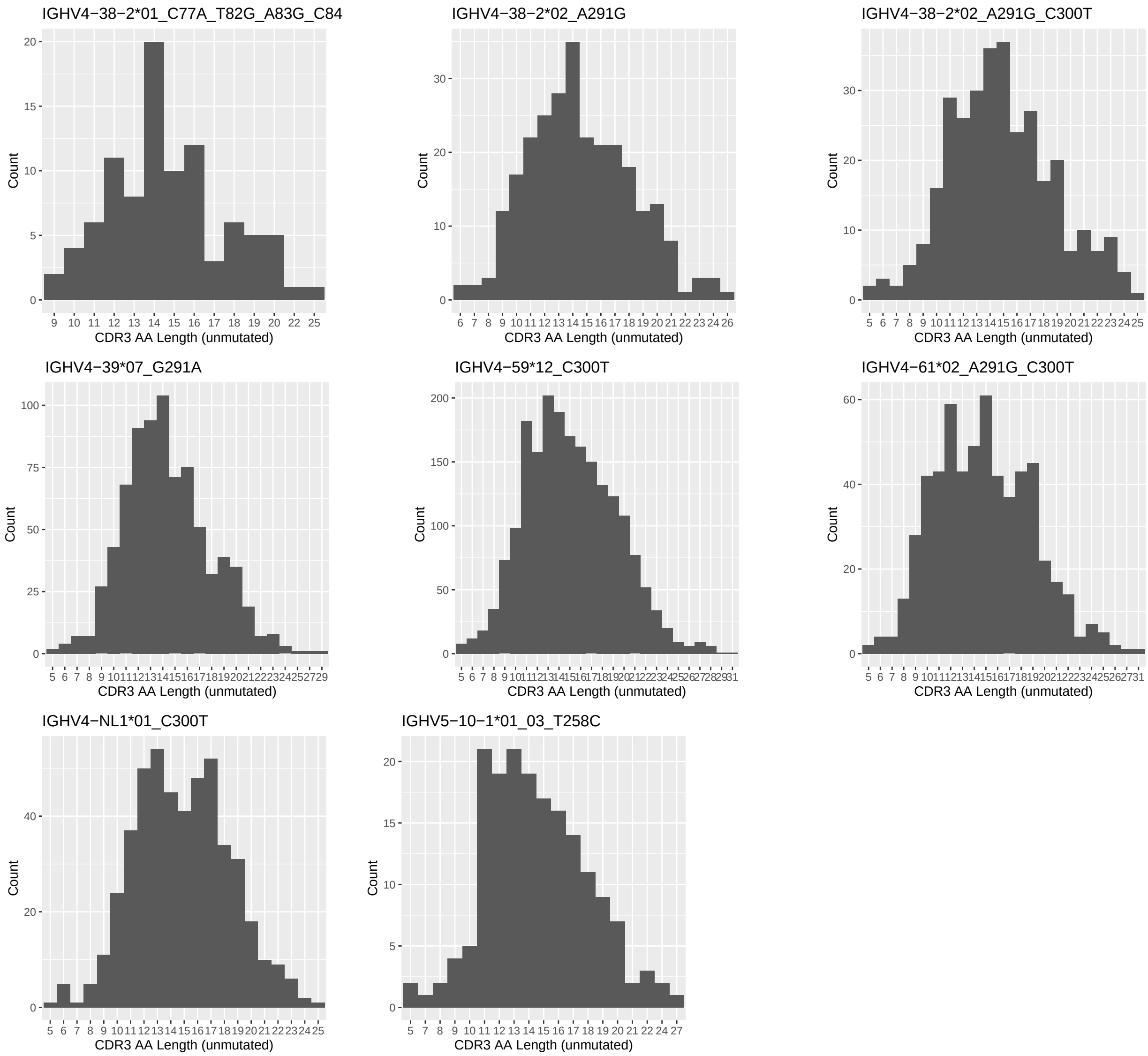


1.5 CDR3 length distribution, in assignments to novel alleles

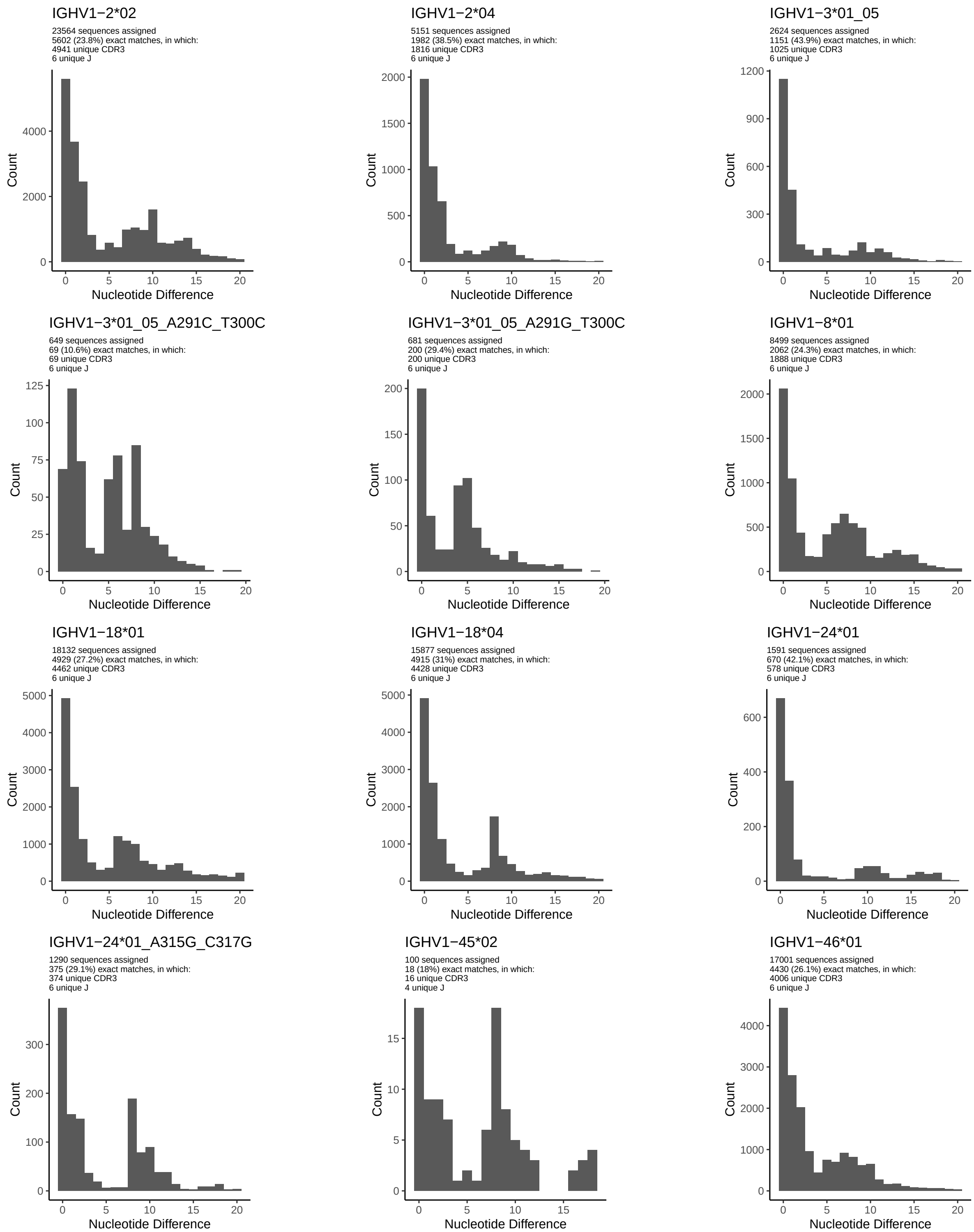


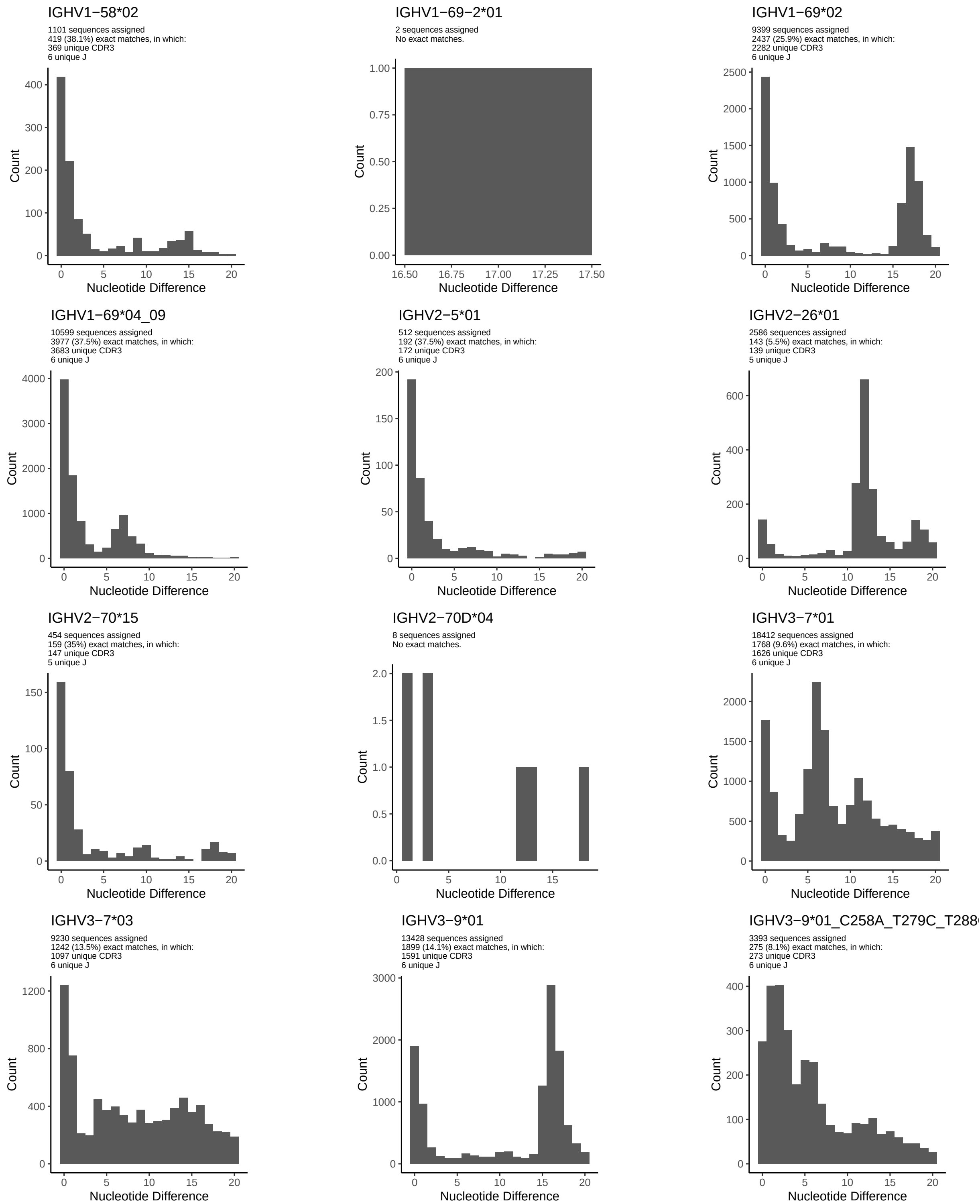

IGHV3−30*03_T246C_T247G_A256T_G2
Count
CDR3 AA Length (unmutated)

IGHV3−30−3*01_T288C
Count
CDR3 AA Length (unmutated)

IGHV3−33*06_C288T
Count
CDR3 AA Length (unmutated)

IGHV3−48*01_G247T_T256A_A258G_T3
Count
CDR3 AA Length (unmutated)

IGHV3−53*01_02_T267G_G303A_G317A
Count
CDR3 AA Length (unmutated)

IGHV3−53*01_02_G81A_G88T_C90T_T1
Count
CDR3 AA Length (unmutated)

IGHV3−64D*09_T314C
Count
CDR3 AA Length (unmutated)

IGHV3−64D*09_T314C_A317G
Count
CDR3 AA Length (unmutated)

IGHV3−66*01_G88T_C90T_T105C_A109
Count
CDR3 AA Length (unmutated)

IGHV3−66*02_T267G
Count
CDR3 AA Length (unmutated)

IGHV3−74*01_A315G
Count
CDR3 AA Length (unmutated)

IGHV3−74*01_C246T_G247T_T279C_C2
Count
CDR3 AA Length (unmutated)

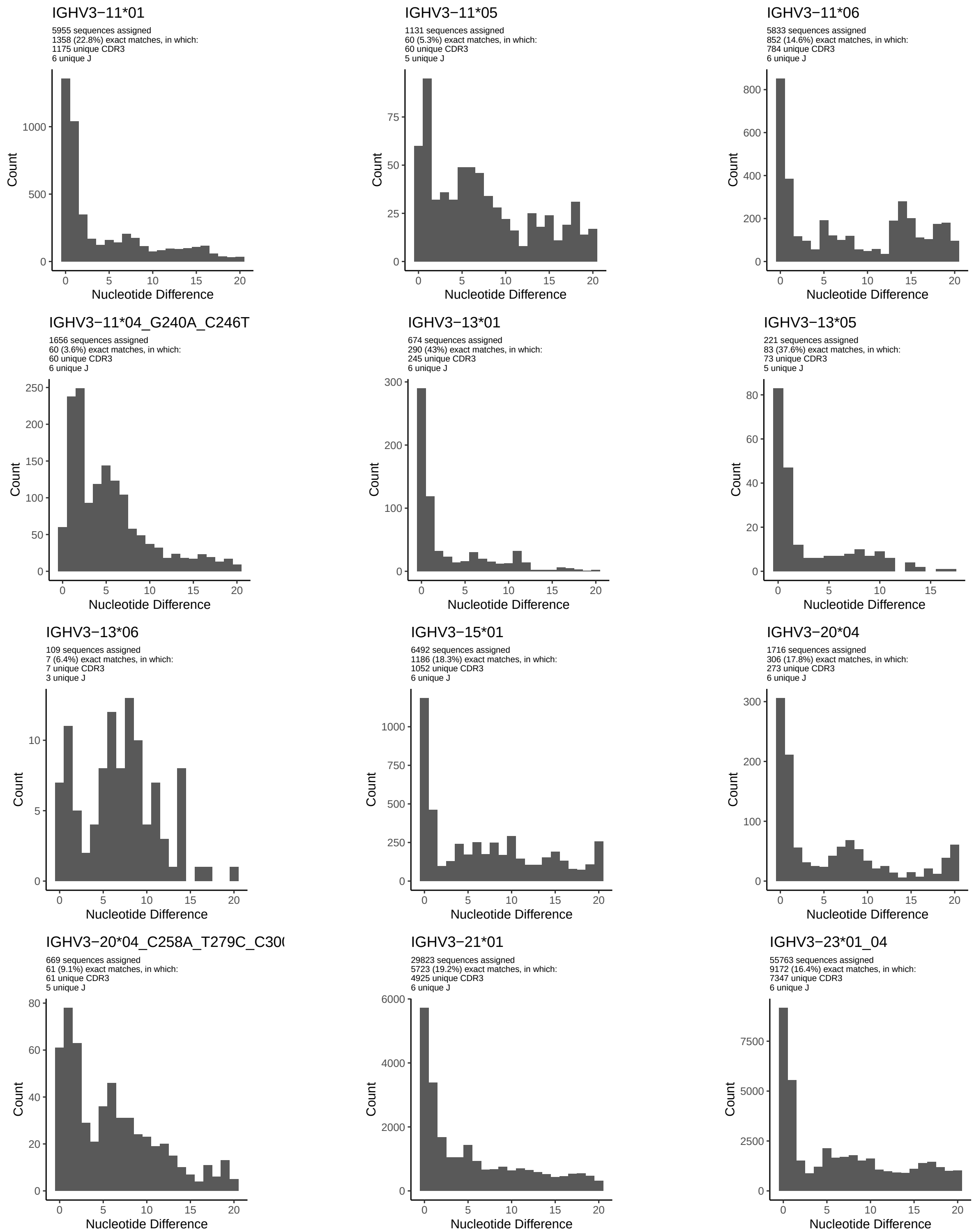

IGHV3−74*01_A256T_G258A_T279C_A31
IGHV4−4*02_03_A245C
IGHV4−4*08_T163C_A164G_C237A
IGHV4−30−2*01_C315G
IGHV4−30−2*01_G245C_C300T_C315G
IGHV4−30−2*01_G245C_G291A_C315G
IGHV4−30−4*01_T285C_A291G_C315G
IGHV4−30−4*01_T285C_C315G
IGHV4−30−4*01_T285C_A291G_C300T_C
IGHV4−31*03_T285C
IGHV4−34*01_T300C
IGHV4−34*01_G291A_T300C
Count
CDR3 AA Length (unmutated)

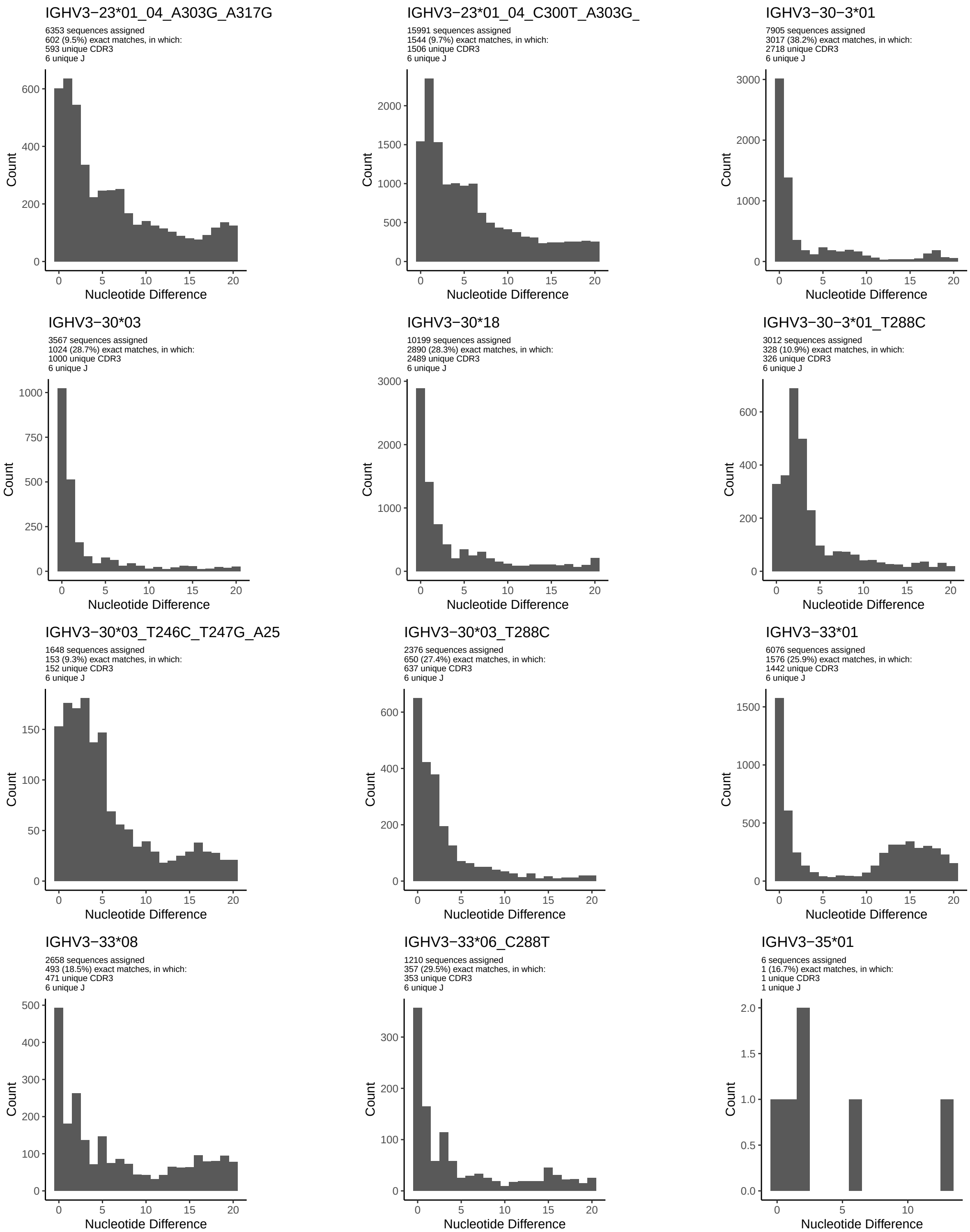

IGHV4−38−2*01_C77A_T82G_A83G_C84
IGHV4−38−2*02_A291G
IGHV4−38−2*02_A291G_C300T
IGHV4−39*07_G291A
IGHV4−59*12_C300T
IGHV4−61*02_A291G_C300T
IGHV4−NL1*01_C300T
IGHV5−10−1*01_03_T258C
Count
CDR3 AA Length (unmutated)

2 Variation from germline, in assignments to each allele

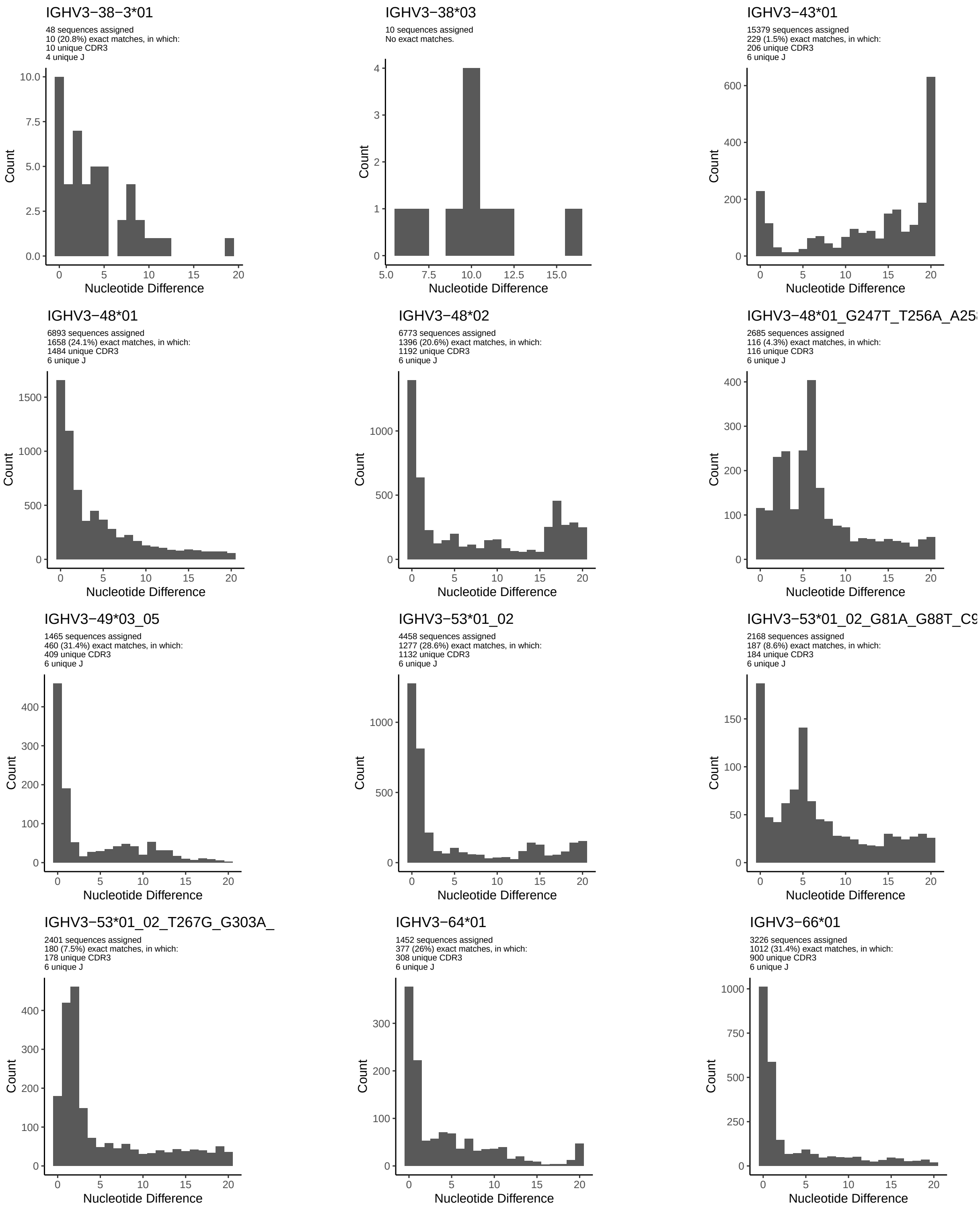


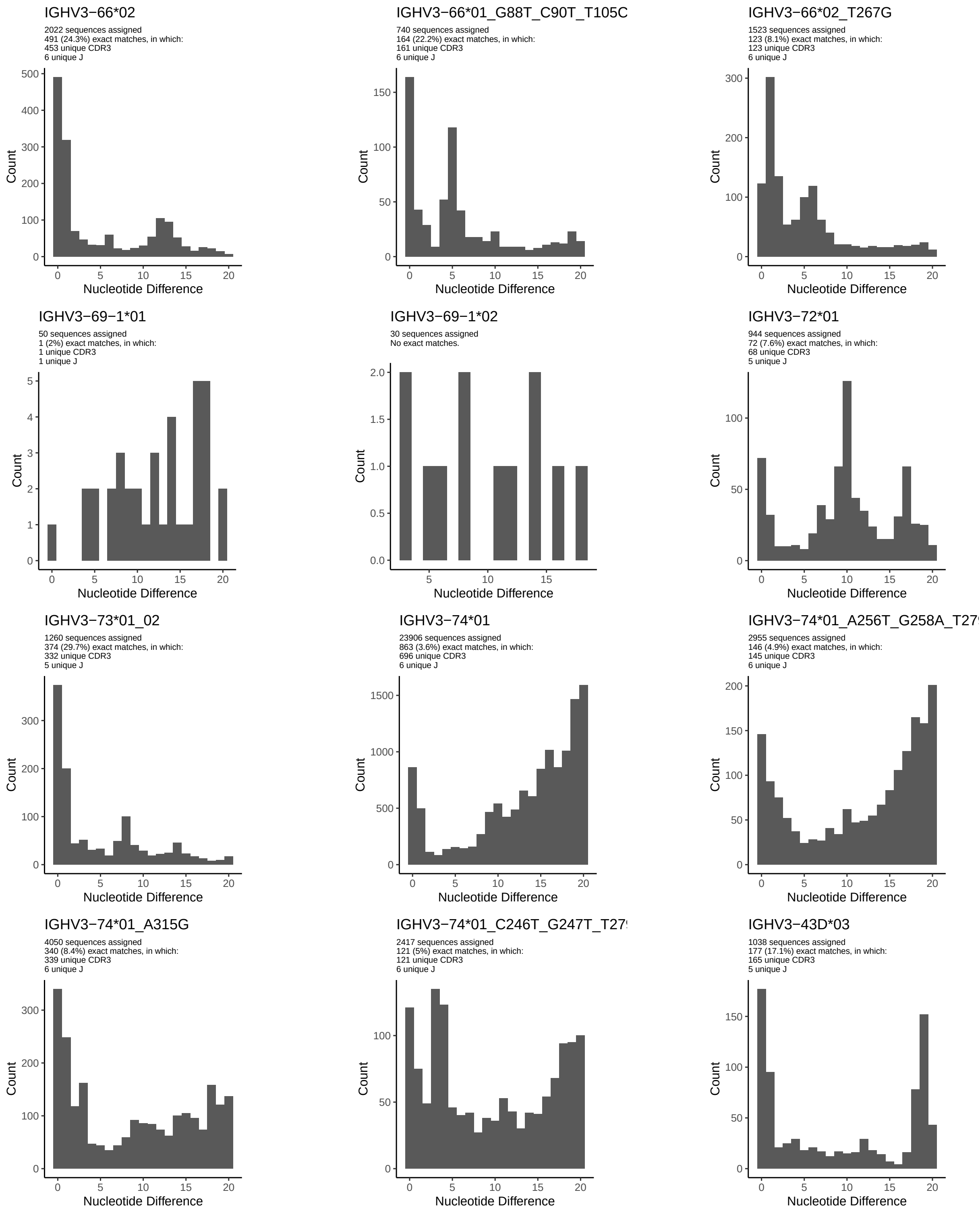

IGHV1−58*02
1101 sequences assigned
419 (38.1%) exact matches, in which:
369 unique CDR3
6 unique J
IGHV1−69−2*01
2 sequences assigned
No exact matches.
IGHV1−69*02
9399 sequences assigned
2437 (25.9%) exact matches, in which:
2282 unique CDR3
6 unique J
IGHV1−69*04_09
10599 sequences assigned
3977 (37.5%) exact matches, in which:
3683 unique CDR3
6 unique J
IGHV2−5*01
512 sequences assigned
192 (37.5%) exact matches, in which:
172 unique CDR3
6 unique J
IGHV2−26*01
2586 sequences assigned
143 (5.5%) exact matches, in which:
139 unique CDR3
5 unique J
IGHV2−70*15
454 sequences assigned
159 (35%) exact matches, in which:
147 unique CDR3
5 unique J
IGHV2−70D*04
8 sequences assigned
No exact matches.
IGHV3−7*01
18412 sequences assigned
1768 (9.6%) exact matches, in which:
1626 unique CDR3
6 unique J
IGHV3−7*03
9230 sequences assigned
1242 (13.5%) exact matches, in which:
1097 unique CDR3
6 unique J
IGHV3−9*01
13428 sequences assigned
1899 (14.1%) exact matches, in which:
1591 unique CDR3
6 unique J
IGHV3−9*01_C258A_T279C_T288
3393 sequences assigned
275 (8.1%) exact matches, in which:
273 unique CDR3
6 unique J
Count
Nucleotide Difference

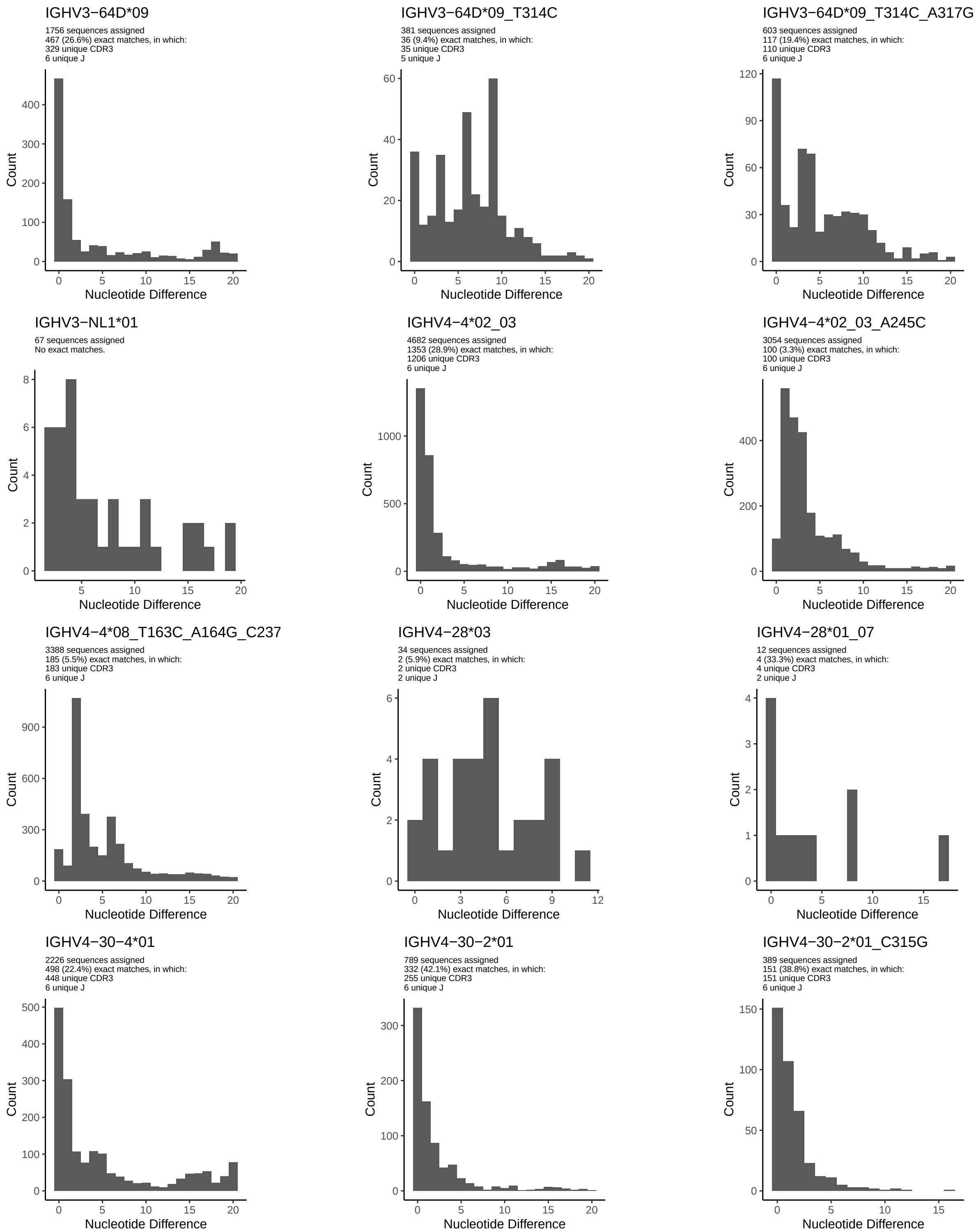

IGHV3−11*01
5955 sequences assigned
1358 (22.8%) exact matches, in which:
1175 unique CDR3
6 unique J
IGHV3−11*05
1131 sequences assigned
60 (5.3%) exact matches, in which:
60 unique CDR3
5 unique J
IGHV3−11*06
5833 sequences assigned
852 (14.6%) exact matches, in which:
784 unique CDR3
6 unique J
IGHV3−11*04_G240A_C246T
1656 sequences assigned
60 (3.6%) exact matches, in which:
60 unique CDR3
6 unique J
IGHV3−13*01
674 sequences assigned
290 (43%) exact matches, in which:
245 unique CDR3
6 unique J
IGHV3−13*05
221 sequences assigned
83 (37.6%) exact matches, in which:
73 unique CDR3
5 unique J
IGHV3−13*06
109 sequences assigned
7 (6.4%) exact matches, in which:
7 unique CDR3
3 unique J
IGHV3−15*01
6492 sequences assigned
1186 (18.3%) exact matches, in which:
1052 unique CDR3
6 unique J
IGHV3−20*04
1716 sequences assigned
306 (17.8%) exact matches, in which:
273 unique CDR3
6 unique J
IGHV3−20*04_C258A_T279C_C30(
669 sequences assigned
61 (9.1%) exact matches, in which:
61 unique CDR3
5 unique J
IGHV3−21*01
29823 sequences assigned
5723 (19.2%) exact matches, in which:
4925 unique CDR3
6 unique J
IGHV3−23*01_04
55763 sequences assigned
9172 (16.4%) exact matches, in which:
7347 unique CDR3
6 unique J
Count
Nucleotide Difference

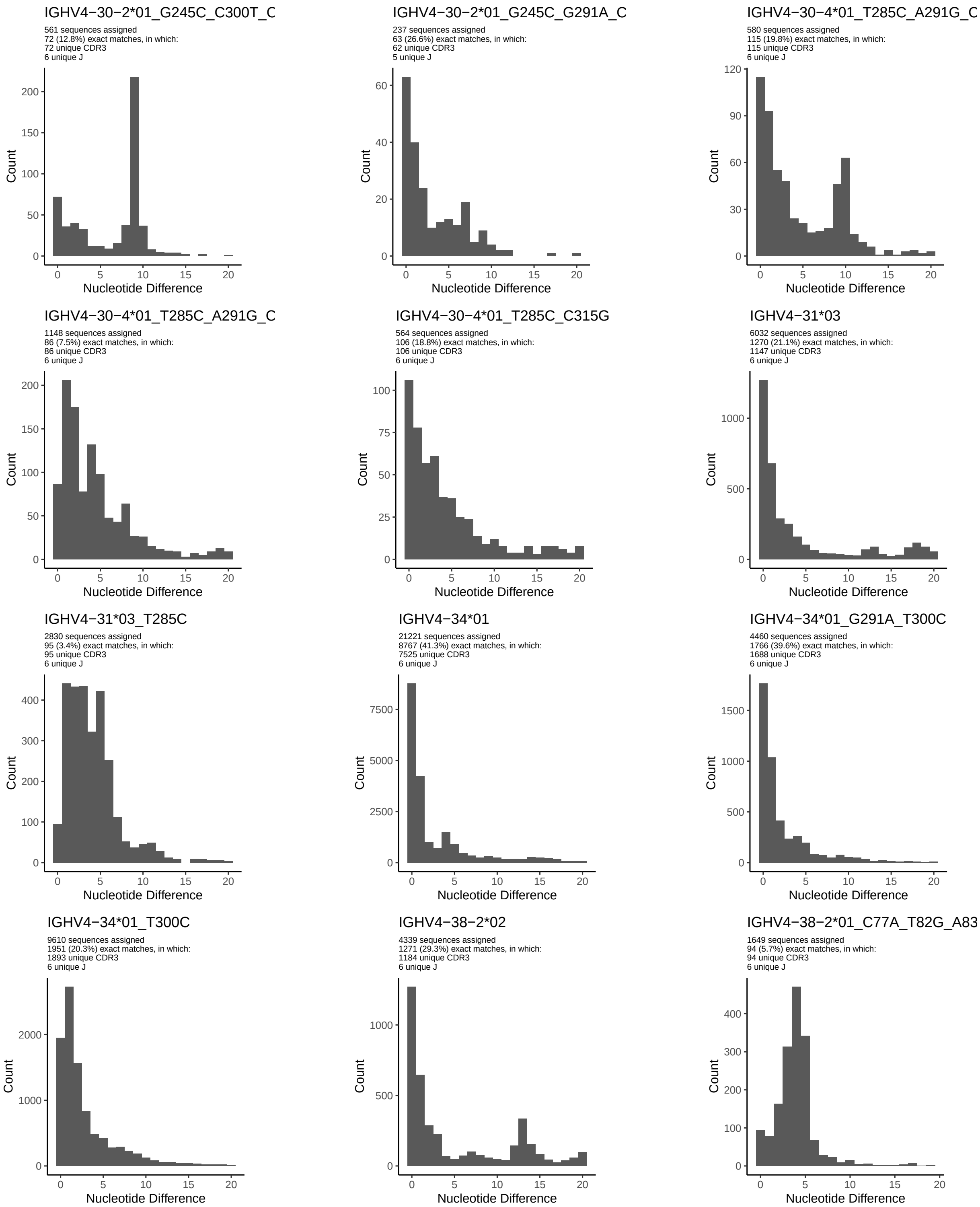


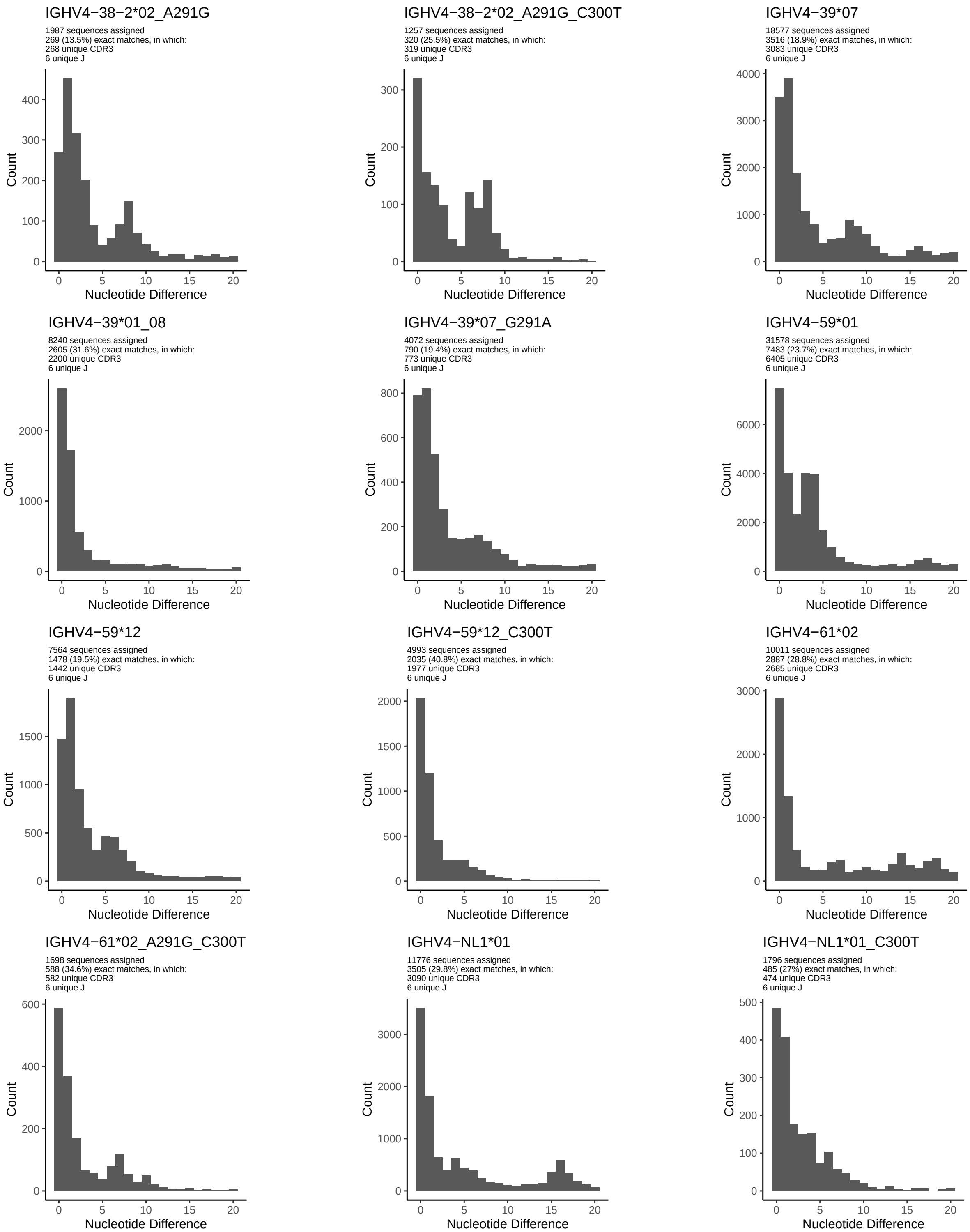
IGHV3−23*01_04_A303G_A317G
6353 sequences assigned
602 (9.5%) exact matches, in which:
593 unique CDR3
6 unique J
IGHV3−23*01_04_C300T_A303G_
15991 sequences assigned
1544 (9.7%) exact matches, in which:
1506 unique CDR3
6 unique J
IGHV3−30−3*01
7905 sequences assigned
3017 (38.2%) exact matches, in which:
2718 unique CDR3
6 unique J
IGHV3−30*03
3567 sequences assigned
1024 (28.7%) exact matches, in which:
1000 unique CDR3
6 unique J
IGHV3−30*18
10199 sequences assigned
2890 (28.3%) exact matches, in which:
2489 unique CDR3
6 unique J
IGHV3−30−3*01_T288C
3012 sequences assigned
328 (10.9%) exact matches, in which:
326 unique CDR3
6 unique J
IGHV3−30*03_T246C_T247G_A25
1648 sequences assigned
153 (9.3%) exact matches, in which:
152 unique CDR3
6 unique J
IGHV3−30*03_T288C
2376 sequences assigned
650 (27.4%) exact matches, in which:
637 unique CDR3
6 unique J
IGHV3−33*01
6076 sequences assigned
1576 (25.9%) exact matches, in which:
1442 unique CDR3
6 unique J
IGHV3−33*08
2658 sequences assigned
493 (18.5%) exact matches, in which:
471 unique CDR3
6 unique J
IGHV3−33*06_C288T
1210 sequences assigned
357 (29.5%) exact matches, in which:
353 unique CDR3
6 unique J
IGHV3−35*01
6 sequences assigned
1 (16.7%) exact matches, in which:
1 unique CDR3
1 unique J
Count
Nucleotide Difference

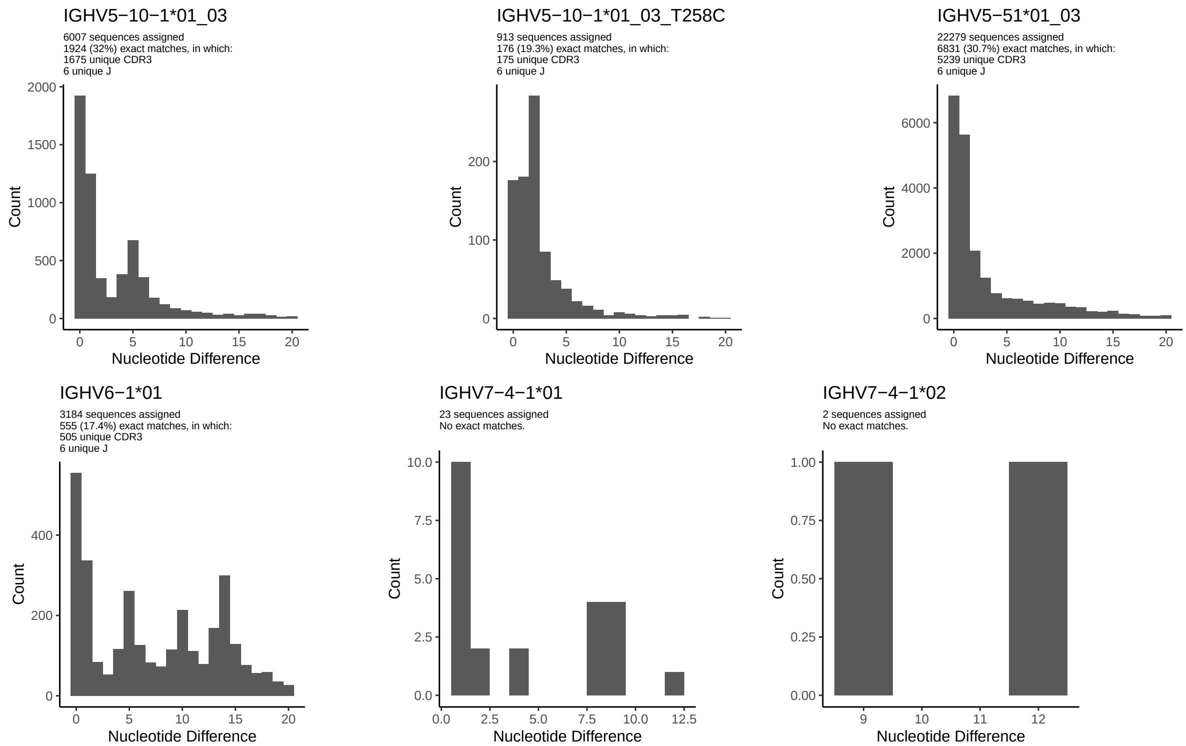

IGHV3−38−3*01
48 sequences assigned
10 (20.8%) exact matches, in which:
10 unique CDR3
4 unique J
IGHV3−38*03
10 sequences assigned
No exact matches.
IGHV3−43*01
15379 sequences assigned
229 (1.5%) exact matches, in which:
206 unique CDR3
6 unique J
IGHV3−48*01
6893 sequences assigned
1658 (24.1%) exact matches, in which:
1484 unique CDR3
6 unique J
IGHV3−48*02
6773 sequences assigned
1396 (20.6%) exact matches, in which:
1192 unique CDR3
6 unique J
IGHV3−48*01_G247T_T256A_A25
2685 sequences assigned
116 (4.3%) exact matches, in which:
116 unique CDR3
6 unique J
IGHV3−49*03_05
1465 sequences assigned
460 (31.4%) exact matches, in which:
409 unique CDR3
6 unique J
IGHV3−53*01_02
4458 sequences assigned
1277 (28.6%) exact matches, in which:
1132 unique CDR3
6 unique J
IGHV3−53*01_02_G81A_G88T_C
2168 sequences assigned
187 (8.6%) exact matches, in which:
184 unique CDR3
6 unique J
IGHV3−53*01_02_T267G_G303A_
2401 sequences assigned
180 (7.5%) exact matches, in which:
178 unique CDR3
6 unique J
IGHV3−64*01
1452 sequences assigned
377 (26%) exact matches, in which:
308 unique CDR3
6 unique J
IGHV3−66*01
3226 sequences assigned
1012 (31.4%) exact matches, in which:
900 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV3−66*02
2022 sequences assigned
491 (24.3%) exact matches, in which:
453 unique CDR3
6 unique J
IGHV3−66*01_G88T_C90T_T105C
740 sequences assigned
164 (22.2%) exact matches, in which:
161 unique CDR3
6 unique J
IGHV3−66*02_T267G
1523 sequences assigned
123 (8.1%) exact matches, in which:
123 unique CDR3
6 unique J
IGHV3−69−1*01
50 sequences assigned
1 (2%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−69−1*02
30 sequences assigned
No exact matches.
IGHV3−72*01
944 sequences assigned
72 (7.6%) exact matches, in which:
68 unique CDR3
5 unique J
IGHV3−73*01_02
1260 sequences assigned
374 (29.7%) exact matches, in which:
332 unique CDR3
5 unique J
IGHV3−74*01
23906 sequences assigned
863 (3.6%) exact matches, in which:
696 unique CDR3
6 unique J
IGHV3−74*01_A256T_G258A_T27
2955 sequences assigned
146 (4.9%) exact matches, in which:
145 unique CDR3
6 unique J
IGHV3−74*01_A315G
4050 sequences assigned
340 (8.4%) exact matches, in which:
339 unique CDR3
6 unique J
IGHV3−74*01_C246T_G247T_T27
2417 sequences assigned
121 (5%) exact matches, in which:
121 unique CDR3
6 unique J
IGHV3−43D*03
1038 sequences assigned
177 (17.1%) exact matches, in which:
165 unique CDR3
5 unique J
Count
Nucleotide Difference

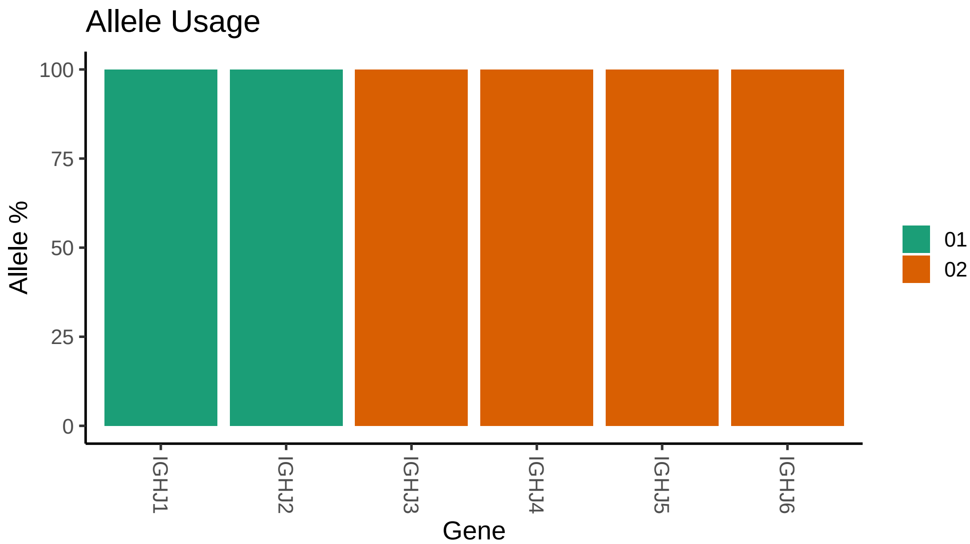

IGHV3−64D*09
1756 sequences assigned
467 (26.6%) exact matches, in which:
329 unique CDR3
6 unique J
IGHV3−64D*09_T314C
381 sequences assigned
36 (9.4%) exact matches, in which:
35 unique CDR3
5 unique J
IGHV3−64D*09_T314C_A317G
603 sequences assigned
117 (19.4%) exact matches, in which:
110 unique CDR3
6 unique J
IGHV3−NL1*01
67 sequences assigned
No exact matches.
IGHV4−4*02_03
4682 sequences assigned
1353 (28.9%) exact matches, in which:
1206 unique CDR3
6 unique J
IGHV4−4*02_03_A245C
3054 sequences assigned
100 (3.3%) exact matches, in which:
100 unique CDR3
6 unique J
IGHV4−4*08_T163C_A164G_C237
3388 sequences assigned
185 (5.5%) exact matches, in which:
183 unique CDR3
6 unique J
IGHV4−28*03
34 sequences assigned
2 (5.9%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV4−28*01_07
12 sequences assigned
4 (33.3%) exact matches, in which:
4 unique CDR3
2 unique J
IGHV4−30−4*01
2226 sequences assigned
498 (22.4%) exact matches, in which:
448 unique CDR3
6 unique J
IGHV4−30−2*01
789 sequences assigned
332 (42.1%) exact matches, in which:
255 unique CDR3
6 unique J
IGHV4−30−2*01_C315G
389 sequences assigned
151 (38.8%) exact matches, in which:
151 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−30−2*01_G245C_C300T_C
561 sequences assigned
72 (12.8%) exact matches, in which:
72 unique CDR3
6 unique J
IGHV4−30−2*01_G245C_G291A_C
237 sequences assigned
63 (26.6%) exact matches, in which:
62 unique CDR3
5 unique J
IGHV4−30−4*01_T285C_A291G_C
580 sequences assigned
115 (19.8%) exact matches, in which:
115 unique CDR3
6 unique J
IGHV4−30−4*01_T285C_A291G_C
1148 sequences assigned
86 (7.5%) exact matches, in which:
86 unique CDR3
6 unique J
IGHV4−30−4*01_T285C_C315G
564 sequences assigned
106 (18.8%) exact matches, in which:
106 unique CDR3
6 unique J
IGHV4−31*03
6032 sequences assigned
1270 (21.1%) exact matches, in which:
1147 unique CDR3
6 unique J
IGHV4−31*03_T285C
2830 sequences assigned
95 (3.4%) exact matches, in which:
95 unique CDR3
6 unique J
IGHV4−34*01
21221 sequences assigned
8767 (41.3%) exact matches, in which:
7525 unique CDR3
6 unique J
IGHV4−34*01_G291A_T300C
4460 sequences assigned
1766 (39.6%) exact matches, in which:
1688 unique CDR3
6 unique J
IGHV4−34*01_T300C
9610 sequences assigned
1951 (20.3%) exact matches, in which:
1893 unique CDR3
6 unique J
IGHV4−38−2*02
4339 sequences assigned
1271 (29.3%) exact matches, in which:
1184 unique CDR3
6 unique J
IGHV4−38−2*01_C77A_T82G_A83
1649 sequences assigned
94 (5.7%) exact matches, in which:
94 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−38−2*02_A291G
1987 sequences assigned
269 (13.5%) exact matches, in which:
268 unique CDR3
6 unique J
IGHV4−38−2*02_A291G_C300T
1257 sequences assigned
320 (25.5%) exact matches, in which:
319 unique CDR3
6 unique J
IGHV4−39*07
18577 sequences assigned
3516 (18.9%) exact matches, in which:
3083 unique CDR3
6 unique J
IGHV4−39*01_08
8240 sequences assigned
2605 (31.6%) exact matches, in which:
2200 unique CDR3
6 unique J
IGHV4−39*07_G291A
4072 sequences assigned
790 (19.4%) exact matches, in which:
773 unique CDR3
6 unique J
IGHV4−59*01
31578 sequences assigned
7483 (23.7%) exact matches, in which:
6405 unique CDR3
6 unique J
IGHV4−59*12
7564 sequences assigned
1478 (19.5%) exact matches, in which:
1442 unique CDR3
6 unique J
IGHV4−59*12_C300T
4993 sequences assigned
2035 (40.8%) exact matches, in which:
1977 unique CDR3
6 unique J
IGHV4−61*02
10011 sequences assigned
2887 (28.8%) exact matches, in which:
2685 unique CDR3
6 unique J
IGHV4−61*02_A291G_C300T
1698 sequences assigned
588 (34.6%) exact matches, in which:
582 unique CDR3
6 unique J
IGHV4−NL1*01
11776 sequences assigned
3505 (29.8%) exact matches, in which:
3090 unique CDR3
6 unique J
IGHV4−NL1*01_C300T
1796 sequences assigned
485 (27%) exact matches, in which:
474 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV5−10−1*01_03
6007 sequences assigned
1924 (32%) exact matches, in which:
1675 unique CDR3
6 unique J
IGHV5−10−1*01_03_T258C
913 sequences assigned
176 (19.3%) exact matches, in which:
175 unique CDR3
6 unique J
IGHV5−51*01_03
22279 sequences assigned
6831 (30.7%) exact matches, in which:
5239 unique CDR3
6 unique J
IGHV6−1*01
3184 sequences assigned
555 (17.4%) exact matches, in which:
505 unique CDR3
6 unique J
IGHV7−4−1*01
23 sequences assigned
No exact matches.
IGHV7−4−1*02
2 sequences assigned
No exact matches.
Count
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /work/jenkins/PRJNA491287/M64 046/M64 046/M64 046/M64 046/99/4440fdff91b11f9ba977a2
##
## Germline reference file: /work/jenkins/PRJNA491287/M64 046/M64 046/M64 046/M64 046/99/4440fdff91b11f
##
## Novel allele file: /work/jenkins/PRJNA491287/M64 046/M64 046/M64 046/M64 046/99/4440fdff91b11f9ba977
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 3 01_05: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 3 01_05_A291G_T300C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 3 01_05_A291C_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 24 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_A315G_C317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 9 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 9 01_C258A_T279C_T288C_C300T_T301G_A315G_A317G: replacing with
##
##
## Unexpected N in reference sequence IGHV3 11 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 11 04_G240A_C246T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 20 04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 20 04_C258A_T279C_C300T_T301G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 23 01_04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_A303G_A317G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C300T_A303G_A317G: replacing with gap
```

```
##
##
## Unexpected N in reference sequence IGHV3 30 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T288C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T246C_T247G_A256T_G258A_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 3 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_C288T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 48 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 48 01_G247T_T256A_A258G_T300C_G303A_G317A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 53 01_02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 53 01_02_T267G_G303A_G317A: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 53 01_02_G81A_G88T_C90T_T105C_A109T_C111T_T112G_A113C: replacin
##
##
## Unexpected N in reference sequence IGHV3 64D 09: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 64D 09_T314C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 64D 09_T314C_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 66 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 66 01_G88T_C90T_T105C_A109T_C111T_T112G_A113C: replacing with g
##
##
## Unexpected N in reference sequence IGHV3 66 02: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV3 66 02_T267G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_C246T_G247T_T279C_C288T_A315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A256T_G258A_T279C_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 61 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 08_T163C_A164G_C237A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_G245C_C300T_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_G245C_G291A_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 4 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_T285C_A291G_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_T285C_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_T285C_A291G_C300T_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 31 03: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV4 31 03_T285C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_G291A_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 38 2 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 01_C77A_T82G_A83G_C84G_A88T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 02_A291G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 02_A291G_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 39 07: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 39 07_G291A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 59 12_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 02_A291G_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 NL1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV5 10 1 01_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV5 10 1 01_03_T258C: replacing with gap
```